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CCNA – New Questions Part 9

 February 25th, 2022 [Go to comments](#)

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Question 1

Refer to the exhibit.

```
Switch#show etherchannel summary
[output omitted]
```

Group	Port-channel	Protocol	Ports	
-----+-----+-----+-----				
10	Po10(SU)	LACP	Gi0/0(P)	Gi0/1(P)
20	Po20(SU)	LACP	Gi0/2(P)	Gi0/3(P)

Which two commands when used together create port channel 10? (Choose two)

- A. int range g0/0-1
channel-group 10 mode active
- B. int range g0/0-1
channel-group 10 mode desirable
- C. int range g0/0-1
channel-group 10 mode passive
- D. int range g0/0-1
channel-group 10 mode auto
- E. int range g0/0-1
channel-group 10 mode on

Answer: A C

Question 2

Which type of IPv6 address is similar to a unicast address but is assigned to multiple devices on the same network at the same time?

- A. global unicast address
- B. anycast address
- C. multicast address
- D. link-local address

Answer: B

Explanation

The anycast address is very similar to the multicast address, but packets will be delivered to only one random host, instead of the entire group. Anycast address don't have a specific range, as they are exactly the same as regular unicast addresses. This means that a host has no way to distinguish a unicast from an anycast address when it sends a packet.

Question 3

Which field within the access-request packet is encrypted by RADIUS?

- A. authorized services
- B. authenticator
- C. username
- D. password

Answer: D

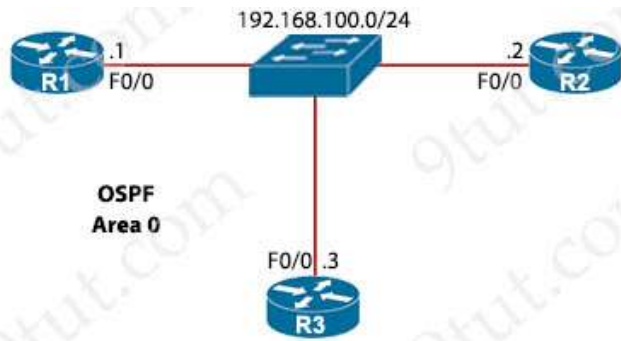
Explanation

RADIUS encrypts only the password in the access-request packet, from the client to the server. The remainder of the packet is unencrypted. Other information, such as username, authorized services, and accounting, can be captured by a third party.

Reference: <https://www.cisco.com/c/en/us/support/docs/security-vpn/remote-authentication-dial-user-service-radius/13838-10.html>

Question 4

Refer to the exhibit.



R1#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.100.2	1	FULL/BDR	00:00:35	192.168.100.2	FastEthernet0/0
192.168.100.3	1	FULL/DR	00:00:34	192.168.100.3	FastEthernet0/0

Which two configurations must the engineer apply on this network so that R1 becomes the DR? (Choose two)

- A. R1(config)#interface fastethernet0/0
R1(config-if)#ip ospf priority 200
- B. R1(config)#router ospf 1
R1(config-router)#router-id 192.168.100.1
- C. R3(config)#interface fastethernet0/0
R3(config-if)#ip ospf priority 0
- D. R1(config)#interface fastethernet0/0
R1(config-if)#ip ospf priority 0
- E. R3(config)#interface fastethernet0/0
R3(config-if)#ip ospf priority 200

Answer: A C

Question 5

Refer to the exhibit.



The router has been configured with a supernet to accommodate the requirement for 380 users on a subnet. The requirement already considers 30% future growth. Which configuration verifies the IP subnet on router R4?

- A. Subnet: 10.7.54.0
Subnet mask: 255.255.254.0
Broadcast address: 10.7.54.255
Usable IP address range: 10.7.54.1 – 10.7.55.254

B. Subnet: 10.7.54.0
Subnet mask: 255.255.254.0
Broadcast address: 10.7.55.255
Usable IP address range: 10.7.54.1 – 10.7.55.254

C. Subnet: 10.7.54.0
Subnet mask: 255.255.128.0
Broadcast address: 10.7.55.255
Usable IP address range: 10.7.54.1 – 10.7.55.254

D. Subnet: 10.7.54.0
Subnet mask: 255.255.255.0
Broadcast address: 10.7.54.255
Usable IP address range: 10.7.54.1 – 10.7.55.254

Answer: B

Explanation

380 users in a subnet $< 512 = 2^9$. Therefore we need 9 bits 0 in the subnet mask $\rightarrow$ 255.255.254.0. The increment is 2 (in the third octet) so the broadcast address is 10.7.55.255.

Question 6

What is a function of a Next-Generation IPS?

- A. makes forwarding decisions based on learned MAC addresses
- B. serves as a controller within a controller-based network
- C. integrates with a RADIUS server to enforce Layer 2 device authentication rules
- D. correlates user activity with network events

Answer: D

Question 7

What is the difference between IPv6 unicast and anycast addressing?

- A. An individual IPv6 unicast address is supported on a single interface on one node but an IPv6 anycast address is assigned to a group of interfaces on multiple nodes.
- B. IPv6 unicast nodes must be explicitly configured to recognize the unicast address, but IPv6 anycast nodes require no special configuration
- C. IPv6 anycast nodes must be explicitly configured to recognize the anycast address, but IPv6 unicast nodes require no special configuration
- D. Unlike an IPv6 anycast address, an IPv6 unicast address is assigned to a group of interfaces on multiple nodes

Answer: A

Question 8

Refer to the exhibit.

```
ip domain-name CNAC.com
!
interface GigabitEthernet0/0/0
 ip address 192.168.1.10 255.255.255.0
 duplex auto
 speed auto
!
line vty 0 15
 login local
```

```
R1#show crypto key mypubkey rsa
```

```
R1#show ssh
```

```
%No SSHv2 server connections running.
%No SSHv1 server connections running.
```

Which two commands must be added to update the configuration of router R1 so that it accepts only encrypted connections? (Choose two)

- A. username CNAC secret R!41!4319115@
- B. crypto key generate rsa 1024
- C. ip ssh version 2
- D. line vty 0 4
- E. transport input ssh

Answer: B E

Explanation

“Encrypted connections” here imply SSH, not Telnet so this question wants to ask to only accept SSH connections.

From the output of “show crypto key mypubkey rsa” command, we learn that current there is no RSA key generated so answer B is surely correct. This command is necessary to generate the RSA key. Although this answer is missing the “modulus” keyword (the full command should be “crypto key generate rsa **modulus** 1024”) but maybe it is just a typo.

Although answer A is also necessary but we are not sure if any username has been configured or not so it is not totally correct.

Question 9

Which action is taken by the data plane within a network device?

- A. looks up an egress interface in the forwarding information base
- B. constructs a routing table based on a routing protocol
- C. provides CLI access to the network device
- D. forwards traffic to the next hop

Answer: D

Question 10

R1 as an NTP server must have:

- * NTP authentication enabled
- * NTP packets sourced from Interface loopback 0
- * NTP stratum 2
- * NTP packets only permitted to client IP 209.165.200.225

How should R1 be configured?

Option A <pre>ntp authenticate ntp authentication-key 2 sha1 CISCO123 ntp source Loopback0 ntp access-group server-only 10 ntp master 2 ! access-list 10 permit udp host 209.165.200.225 any eq 123</pre>	Option B <pre>ntp authenticate ntp authentication-key 2 md5 CISCO123 ntp source Loopback0 ntp access-group server-only 10 ntp stratum 2 ! access-list 10 permit udp host 209.165.200.225 any eq 123</pre>
Option C <pre>ntp authenticate ntp authentication-key 2 md5 CISCO123 ntp interface Loopback0 ntp access-group server-only 10 ntp stratum 2 ! access-list 10 permit 209.165.200.225</pre>	Option D <pre>ntp authenticate ntp authentication-key 2 md5 CISCO123 ntp source Loopback0 ntp access-group server-only 10 ntp master 2 ! access-list 10 permit 209.165.200.225</pre>

- A. Option A
- B. Option B
- C. Option C
- D. Option D

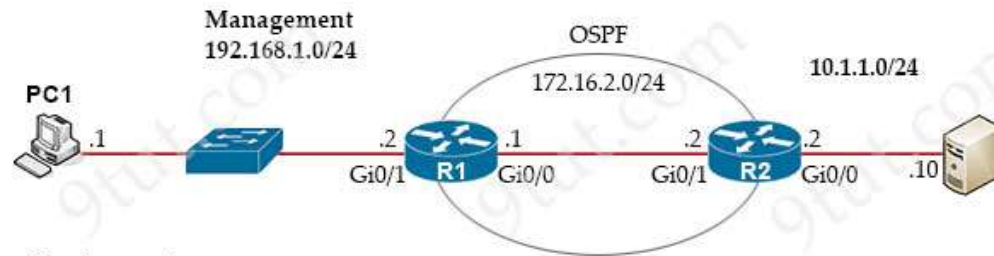
Answer: D

Explanation

To configure a Cisco device as an Authoritative NTP Server, use the **ntp master** [*stratum*] command -> Only Option A and Option D are correct.
 ACL 10 is a standard ACL so we cannot configure protocol (UDP in this case) for the ACL. We can only configure source IP address -> Option D is correct.

Question 11

Refer to the exhibit.



During outage

```
R1#show ip route 10.1.1.10
% Network not in table
```

Normal operation

```
R1#show ip route 10.1.1.10
Routing entry for 10.1.1.0/24
Known via "ospf 1", distance 110, metric 2, type intra area
Last update from 172.16.2.2 on GigabitEthernet0/0, 00:00:18 ago
Routing Descriptor Blocks:
* 172.16.2.2, from 10.1.1.10, 00:00:18 ago, via GigabitEthernet0/0
Route metric is 2, traffic share count is 1
```

Which route must be configured on R1 so that OSPF routing is used when OSPF is up. But the server is still reachable when OSPF goes down?

- A. ip route 10.1.1.10 255.255.255.255 172.16.2.2 100
- B. ip route 10.1.1.0 255.255.255.0 gi0/1 125
- C. ip route 10.1.1.0 255.255.255.0 172.16.2.2 100
- D. ip route 10.1.1.10 255.255.255.255 gi0/0 125

Answer: D

Question 12

How does Rapid PVST+ create a fast loop-free network topology?

- A. It requires multiple links between core switches
- B. It maps multiple VLANs into the same spanning-tree instance

- C. It generates one spanning-tree instance for each VLAN
- D. It uses multiple active paths between end stations

Answer: C

Question 13

Which WLC management connection type is vulnerable to man-in-the-middle attacks?

- A. SSH
- B. HTTPS
- C. Telnet
- D. console

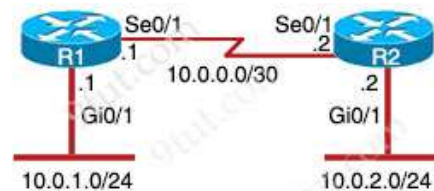
Answer: C

Explanation

As you know, telnet is insecure. By default, telnet is disabled on Cisco WLCs. So, if you want to use telnet, you must enable it.

Question 14

Refer to the exhibit.



Which command configures OSPF on the point-to-point link between routers R1 and R2?

- A. network 10.0.0.0 0.0.0.255 area 0
- B. neighbor 10.1.2.0 cost 180
- C. ip ospf priority 100
- D. router-id 10.0.0.15

Answer: A

Question 15

Which characteristic differentiates the concept of authentication from authorization and accounting?

- A. user-activity logging
- B. service limitations
- C. consumption-based billing
- D. identity verification

Answer: D

Question 16

(this question is duplicated so we removed it)

Question 17

Refer to the exhibit.

TenGigabitEthernet0/0/0 is up, line protocol is up
Hardware is BUILT-IN-2T+6X1GE, address is 74a0.2f7a.0123 (bia 74a0.2f7a.0123)
Description: Uplink
Internet address is 10.1.1.1/24
MTU 1500 bytes, BW 10000000 Kbit/sec, DLY 10 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive not supported
Full Duplex, 10000Mbps, link type is force-up, media type is unknown media type
output flow control is on, input flow-control is on
ART type: ARPA, ARP Timeout 04:00:00
Last input 00:00:00, output 00:05:40, output hang never
Last clearing of "show interface" counters never
Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 6160000 bits/sec, 1113 packets/sec
5 minute output rate 11213000 bits/sec, 1553 packets/sec
12662416065 packets input, 12607032232894 bytes, 0 no buffer
Received 14117163 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 watchdog, 26271385 multicast, 0 pause input
7907770090 packets output, 5072790424092 bytes, 0 underruns
0 output errors, 8662414049 collisions, 1 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier, 0 pause output
0 output buffer failures, 0 output buffers swapped out
1 carrier transitions

Traffic that is flowing over interface TenGigabitEthernet0/0 experiences slow transfer speeds. What is the reason for the issue?

- A. heavy traffic congestion
- B. queuing drops
- C. a speed conflict
- D. a duplex incompatibility

Answer: D

Explanation

We see the collisions are very high compared to the packets output. This is usually the result of duplex mismatch (one is half-duplex, one end is full-duplex for example)

Question 18

Which type of network attack overwhelms the target server by sending multiple packets to a port until the half-open TCP resources of the target are exhausted?

- A. SYN flood
- B. reflection
- C. teardrop
- D. amplification

Answer: A

Explanation

A SYN flood (half-open attack) is a type of denial-of-service (DDoS) attack which aims to make a server unavailable to legitimate traffic by consuming all available server resources. By repeatedly sending initial connection request (SYN) packets, the attacker is able to overwhelm all available ports on a targeted server machine, causing the targeted device to respond to legitimate traffic sluggishly or not at all.

Question 19

Which interface mode must be configured to connect the lightweight APs in a centralized architecture?

- A. WLAN dynamic
- B. management
- C. trunk
- D. access

Answer: D

Question 20

Which two network actions occur within the data plane? (Choose two)

- A. Add or remove an 802.1Q trunking header.
- B. Make a configuration change from an incoming NETCONF RPC.
- C. Run routing protocols.
- D. Reply to an incoming ICMP echo request.
- E. Match the destination MAC address to the MAC address table.

Answer: A E

Explanation

For perspective, the following list details some of the more common actions that a networking device does that fit into the data plane:

- + De-encapsulating and re-encapsulating a packet in a data-link frame (routers, Layer 3 switches)
- + **Adding or removing an 802.1Q trunking header** (routers and switches)
- + **Matching an Ethernet frame's destination Media Access Control (MAC) address to the MAC address table** (Layer 2 switches)
- + Matching an IP packet's destination IP address to the IP routing table (routers, Layer 3 switches)
- + Encrypting the data and adding a new IP header (for virtual private network [VPN] processing)
- + Changing the source or destination IP address (for Network Address Translation [NAT] processing)
- + Discarding a message due to a filter (access control lists [ACLs], port security)

Reference: <https://www.ciscopress.com/articles/article.asp?p=2995354&seqNum=2>

Question 21

Refer to the exhibit.

```
A#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
172.1.1.1 1 EXCHANGE/- 00:00:36 172.16.32.1 Serial0.1
```

An engineer assumes a configuration task from a peer. Router A must establish an OSPF neighbor relationship with neighbor 172.1.1.1. The output displays the status of the adjacency after 2 hours. What is the next step in the configuration process for the routers to establish an adjacency?

- A. Set the router B OSPF ID to the same value as its IP address
- B. Set the router B OSPF ID to a nonhost address
- C. Configure a point-to-point link between router A and router B
- D. Configure router A to use the same MTU size as router B

Answer: D

Explanation

The problem occurs most frequently when attempting to run OSPF between a Cisco router and another vendor's router. The problem occurs when the maximum transmission unit (MTU) settings for neighboring router interfaces don't match.

Reference: <https://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/13684-12.html>

Question 22

Refer to the exhibit.

```
CPE#show ip route
192.168.1.0/24 is variably subnetted, 3 subnets, 3 masks
B   192.168.1.0/24 [20/1] via 192.168.12.2, 00:00:06
R   192.168.1.128/25 [120/5] via 192.168.13.3, 00:02:22, Ethernet0/1
O   192.168.1.192/26 [110/11] via 192.168.14.4, 00:02:22, Ethernet0/2
D   192.168.1.224/27 [90/1024640] via 192.168.15.5, 00:01:33, Ethernet0/3
```

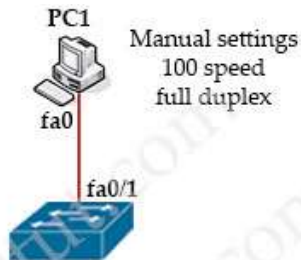
All traffic enters the CPE router from interface Serial0/3 with an IP address of 192.168.50.1. Web traffic from the WAN is destined for a LAN network where servers are load-balanced. An IP packet with a destination address of the HTTP virtual IP of 192.168.1.250 must be forwarded. Which routing table entry does the router use?

- A. 192.168.1.0/24 via 192.168.12.2
- B. 192.168.1.128/25 via 192.168.13.3
- C. 192.168.1.192/26 via 192.168.14.4
- D. 192.168.1.224/27 via 192.168.15.5

Answer: D

Question 23

Refer to the exhibit.



```
Switch#show interface status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	1	auto	auto	10/100BaseTX

The link between PC1 and the switch is up, but it is performing poorly. Which interface condition is causing the performance problem?

- A. There is a duplex mismatch on the interface
- B. There is an issue with the fiber on the switch interface
- C. There is a speed mismatch on the interface
- D. There is an interface type mismatch

Answer: A

Question 24

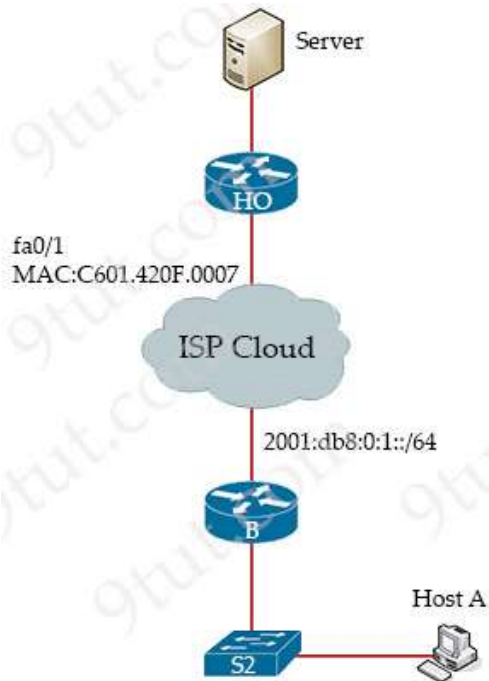
What provides centralized control of authentication and roaming in an enterprise network?

- A. a lightweight access point
- B. a firewall
- C. a wireless LAN controller
- D. a LAN switch

Answer: C

Question 25

Refer to the exhibit.



An engineer is configuring the HO router. Which IPv6 address configuration must be applied to the router fa0/1 interface for the router to assign a unique 64-bit IPv6 address to itself?

- A. ipv6 address 2001:DB8:0:1:C601:42FF:FE0F:7/64
- B. ipv6 address 2001:DB8:0:1:C601:42FE:800F:7/64

- C. ipv6 address 2001:DB8:0:1:FFFF:C601:420F:7/64
- D. ipv6 address 2001:DB8:0:1:FE80:C601:420F:7/64

Answer: A

Explanation

We must use IPv6 EUI-64 so that the router can assign a unique 64-bit IPv6 address to itself. Therefore we get the MAC address of C601.420F.0007 and:

1. Split the MAC address in two halves (6 hex digits each) (so we have C601:42 for the first halves and 0F:0007 for the second halves)
2. Insert FFFE in between the two, making the interface ID.
3. Invert the seventh bit (to 1) of the interface ID if it is 0 (in this question this step is ignored as C6 = 11000110 -> the seventh bit is already 1)

-> We get the interface ID of C601:42FF:FE0F:0007 or C601:42FF:FE0F:7 in short. We prepend the network prefix (64 bits) and get 2001:DB8:0:1:C601:42FF:FE0F:7

Question 26

Refer to the exhibit.



```

Router#show run
Building configuration...
!
interface GigabitEthernet0/0
 ip address 10.10.10.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 172.16.2.1 255.255.255.0
 duplex auto
 speed auto
!

```

An engineer is configuring a new router on the network and applied this configuration. Which additional configuration allows the PC to obtain its IP address from a DHCP server?

- A. Configure the **ip dhcp relay information** command under interface Gi0/1
- B. Configure the **ip dhcp smart-relay** command globally on the router
- C. Configure the **ip helper-address 172.16.2.2** command under interface Gi0/0
- D. Configure the **ip address dhcp** command under interface Gi0/0

Answer: C

Question 27

Refer to the exhibit.



A static route must be configured on R14 to forward traffic for the 172.21.34.0/25 network that resides on R86. Which command must be used to fulfill the request?

- A. ip route 172.21.34.0 255.255.255.192 10.73.65.65
- B. ip route 172.21.34.0 255.255.255.0 10.73.65.65
- C. ip route 172.21.34.0 255.255.128.0 10.73.65.64
- D. ip route 172.21.34.0 255.255.255.128 10.73.65.66

Answer: D

Question 28

What is a function of Opportunistic Wireless Encryption in an environment?

- A. offer compression
- B. increase security by using a WEP connection
- C. provide authentication
- D. protect traffic on open networks

Answer: D

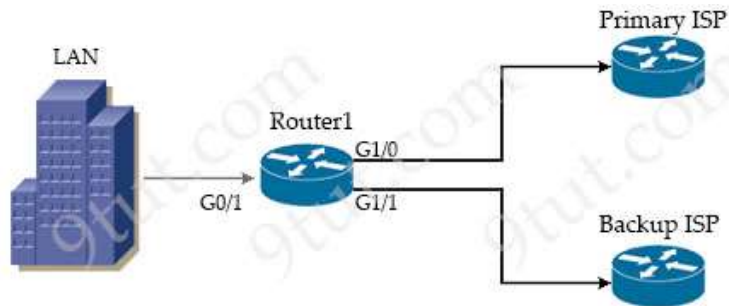
Explanation

Opportunistic Wireless Encryption (OWE) is an extension to IEEE 802.11 that provides encryption of the wireless medium. The purpose of OWE based authentication is avoid open unsecured wireless connectivity between the AP's and clients. The OWE uses the Diffie-Hellman algorithms based Cryptography to setup the wireless encryption.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/9800/16-12/config-guide/b_wl_16_12_cg/wpa3.html

Question 29

Refer to the exhibit.



A company is configuring a failover plan and must implement the default routes in such a way that a floating static route will assume traffic forwarding when the primary link goes down. Which primary route configuration must be used?

- A. ip route 0.0.0.0 0.0.0.0 192.168.0.2 GigabitEthernet1/0
- B. ip route 0.0.0.0 0.0.0.0 192.168.0.2 tracked
- C. ip route 0.0.0.0 0.0.0.0 192.168.0.2 floating
- D. ip route 0.0.0.0 0.0.0.0 192.168.0.2

Answer: D

Question 30

Which action implements physical access control as part of the security program of an organization?

- A. setting up IP cameras to monitor key infrastructure
- B. backing up syslogs at a remote location
- C. configuring enable passwords on network devices
- D. configuring a password for the console port

Answer: A

Explanation

Physical access control can take a number of forms, but the basic idea is to create barriers to prevent unauthorized people from entering a physical space.

Also “camera” is also listed as a physical access control in this Cisco link:

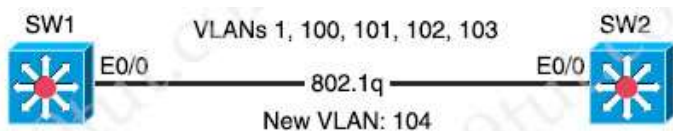
https://www.cisco.com/c/dam/global/hr_hr/assets/images/Cisco_rjesenja_za_zastitu_objekata_i_imovine_-_Alper_Erdal.pdf

Physical Security Systems



Question 31

Refer to the exhibit.



An engineer is asked to insert the new VLAN into the existing trunk without modifying anything previously configured. Which command accomplishes this task?

- A. switchport trunk allowed vlan 100-104
- B. switchport trunk allowed vlan all
- C. switchport trunk allowed vlan add 104
- D. switchport trunk allowed vlan 104

Answer: C

Question 32

Refer to the exhibit.

Hardware is ISR4331-3x1GE, address is 5486.bc25.1f70 (bia 5486.bc25.1f70)

Description: << WAN Link >>

Internet address is 192.0.2.2/30

MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,

reliability 255/255, txload 1/255, rxload 1/255

Encapsulation ARPA, loopback not set

Keepalive not supported

Full Duplex, 1000Mbps, link type is auto, media type is RJ45

output flow control is off, input flow-control is off

ART type: ARPA, ARP Timeout 04:00:00

Last input 00:00:00, output 00:00:11, output hang never

Last clearing of "show interface" counters never

Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0

Queueing strategy: fifo

Output queue: 0/40 (size/max)

5 minute input rate 7000 bits/sec, 4 packets/sec

5 minute output rate 4000 bits/sec, 4 packets/sec

22579370 packets input, 8825545968 bytes, 0 no buffer

Received 67 broadcasts (0 IP multicasts)

0 runs, 0 giants, 0 throttles

3612699 input errors, 3612699 CRC, 0 frame, 0 overrun, 0 ignored

0 watchdog, 10747057 multicast, 0 pause input

12072167 packets output, 1697953637 bytes, 0 underruns

0 output errors, 0 collisions, 1 interface resets

6 unknown protocol drops

0 babbles, 0 late collision, 0 deferred

5 lost carrier, 0 no carrier, 0 pause output

0 output buffer failures, 0 output buffers swapped out

What is a reason for poor performance on the network interface?

- A. The interface is receiving excessive broadcast traffic.
- B. The cable connection between the two devices is faulty.
- C. The interface is operating at a different speed than the connected device.
- D. The bandwidth setting of the interface is misconfigured

Answer: B

Explanation

A working interface (in an up/up state) can still suffer from issues related to the physical cabling as well. The cabling problems might not be bad enough to cause a complete failure, but the transmission failures result in some frames failing to pass successfully over the cable. For example, excessive interference on the cable can cause the various input error counters to keep growing larger, especially the CRC counter. In particular, if the CRC errors grow, but the collisions counters do not, the problem might simply be interference on the cable. (The switch counts each collided frame as one form of input error as well.)

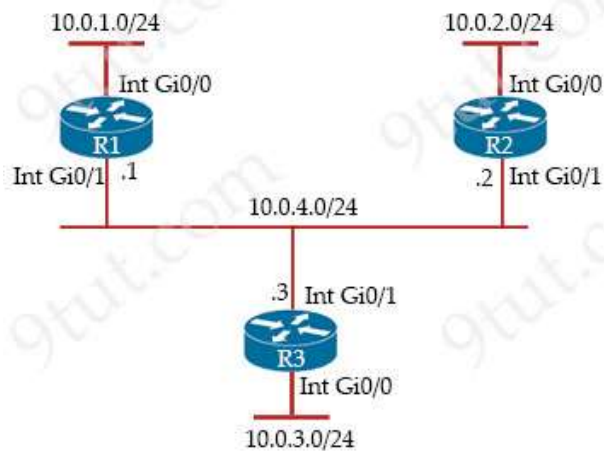
Reference: CCNA 200-301 Official Cert Guide, Volume 1 page 168

In this question, only the input errors and CRC errors grow -> “The cable connected between two devices is faulty” is the best choice.

Note: One problem, called late collisions, points to the classic duplex mismatch problem. But there is no late collision errors so duplex mismatch is not an in this question.

Question 33

Refer to the exhibit.



Routers R1 and R3 have the default configuration. The router R2 priority is set to 99. Which commands on R3 configure it as the DR in the 10.0.4.0/24 network?

- A. R3(config)#interface Gig0/1
R3(config-if)#ip ospf priority 100
- B. R3(config)#interface Gig0/0
R3(config-if)#ip ospf priority 100
- C. R3(config)#interface Gig0/0
R3(config-if)#ip ospf priority 1
- D. R3(config)#interface Gig0/1
R3(config-if)#ip ospf priority 0

Answer: A

Question 34

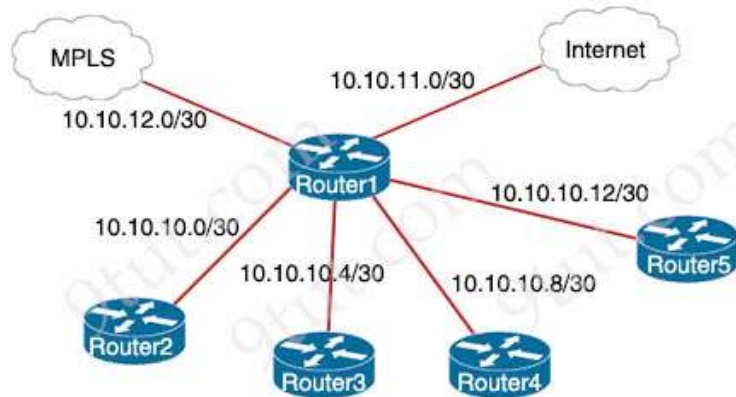
Which QoS per-hop behavior changes the value of the ToS field in the IPv4 packet header?

- A. shaping
- B. marking
- C. policing
- D. classification

Answer: B

Question 35

Refer to the exhibit.



```

Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
  209.165.200.0/27 is subnetted, 1 subnets
B    209.165.200.224 [20/0] via 10.10.12.2,03:32:14
  209.165.201.0/27 is subnetted, 1 subnets
B    209.165.201.0 [20/0] via 10.10.12.2,02:26:53
  209.165.202.0/27 is subnetted, 1 subnets
B    209.165.202.128 [20/0] via 10.10.12.2,02:46:03
  10.0.0.0/8 is variably subnetted, 10 subnets, 4 masks
O    10.10.13.0/25 [110/2] via 10.10.10.1,00:00:04, GigabitEthernet0/0
O    10.10.13.128/28 [110/2] via 10.10.10.5,00:00:12, GigabitEthernet0/1
O    10.10.13.144/28 [110/2] via 10.10.10.9,00:01:57, GigabitEthernet0/2
O    10.10.13.160/29 [110/2] via 10.10.10.5,00:00:12, GigabitEthernet0/1
O    10.10.13.208/29 [110/2] via 10.10.10.13,00:01:57, GigabitEthernet0/3
S*  0.0.0.0/0 [1/0] via 10.10.11.2
  
```

Which next-hop IP address does Router1 use for packets destined to host 10.10.13.158?

- A. 10.10.10.5
- B. 10.10.11.2
- C. 10.10.12.2
- D. 10.10.10.9

Answer: D

Explanation

10.10.13.158 belongs to prefix 10.10.13.144/28 so the next-hop 10.10.10.9 will be chosen to forward packet for this destination.

Question 36

What is one reason to implement LAG on a Cisco WLC?

- A. to increase security and encrypt management frames
- B. to provide link redundancy and load balancing
- C. to allow for stateful and link-state failover
- D. to enable connected switch ports to failover and use different VLANs

Answer: B

Question 37

Refer to the exhibit.

R1# show ip route

Codes:

C - connected, S - static, I - IGRP, R - RIP, M - mobile, B — BGP, D -

EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA

external type 1, N2 - OSPF NSSA external type 2, E1 — OSPF external type 1,

E2 - OSPF external type 2, E - EGP

i - IS-IS, LI - IS-IS level-1, L2 - IS-IS level-2, * - candidate default,

U - per-user static route, o- ODR

Gateway of last resort not set

- C 10.0.0.0/8 is directly connected, Loopback0
- 10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
- O 10.0.1.3/32 [110/100] via 10.0.1.100, 00:39:08, Serial0
- C 10.0.1.0/24 is directly connected, Serial0
- O 10.0.1.5/32 [110/5] via 10.0.1.50, 00:39:08, Serial0
- O 10.0.10.0/24 [110/10] via 10.0.1.4, 00:39:08, Gigabit Ethernet 0/0
- D 10.0.10.0/24 [90/10] via 10.0.1.5, 00:39:08, Gigabit Ethernet 0/1

Web traffic is coming in from the WAN interface. Which route takes precedence when the router is processing traffic destined for the LAN network at 10.0.10.0/24?

- A. via next-hop 10.0.1.5
- B. via next-hop 10.0.1.4
- C. via next-hop 10.0.1.50
- D. via next-hop 10.0.1.100

Answer: A

Question 38

Which PoE mode enables powered-device detection and guarantees power when the device is detected?

- A. dynamic
- B. static
- C. active
- D. auto

Answer: B

Explanation

static—Enables powered-device detection. Pre-allocate (reserve) power for a port before the switch discovers the powered device. The switch reserves power for this port even when no device is connected and guarantees that power will be provided upon device detection.

Reference:

https://www.cisco.com/en/US/docs/switches/lan/catalyst3850/software/release/3.2_0_se/multibook/configuration_guide/b_consolidated_config_guide_3850_chapter_011010.html

Note: “auto” – Enables powered-device detection; if enough power is available, automatically allocates power to the PoE port after device detection (default setting).” The answer “auto” is not correct as “auto” only allocates power “if enough power is available”.

Question 39

A Cisco engineer must configure a single switch interface to meet these requirements

- * accept untagged frames and place them in VLAN 20
- * accept tagged frames in VLAN 30 when CDP detects a Cisco IP phone

Which command set must the engineer apply?

- A. switchport mode access
switchport access vlan 20
switchport voice vlan 30
- B. switchport mode trunk
switchport access vlan 20
switchport voice vlan 30
- C. switchport mode dynamic auto
switchport trunk native vlan 20
switchport trunk allowed vlan 30
switchport voice vlan 30
- D. switchport mode dynamic desirable
switchport access vlan 20
switchport trunk allowed vlan 30
switchport voice vlan 30

Answer: A

Explanation

With the configuration in answer:

```
“switchport mode access  
switchport access vlan 20  
switchport voice vlan 30”
```

This configuration is typically used when connecting a Cisco IP phone and a PC to the same switch port.

- + The PC sends untagged traffic -> assigned to VLAN 20.
- + The IP Phone sends 802.1Q-tagged voice traffic -> placed in VLAN 30.
- + CDP helps the switch identify the IP phone and treat its traffic appropriately.

Note:

+ Trunks are used between switches or for special cases, not typical for connecting end devices like phones and PCs simultaneously -> Other answers with “trunk” mode are not correct.

Question 40

Refer to the exhibit.

```
Router#show run  
Building configuration...  
Current configuration : 1530 bytes  
!  
Last configuration change at 11:32:53 UTC Thu Feb 10 2020  
upgrade ipd auto  
version 15.2  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
!  
hostname Router  
!  
boot-start-marker  
boot-end-marker  
!  
!  
!  
no aaa new-model  
no ip icmp rate-limit unreachable  
!  
!  
--More--
```

Which minimum configuration items are needed to enable Secure Shell version 2 access to R15?

A. Router(config)#hostname R15
R15(config)#crypto key generate rsa general-keys modulus 1024
R15(config-line)#line vty 0 15
R15(config-line)# transport input ssh
R15(config)#ip ssh source-interface Fa0/0
R15(config)#ip ssh stricthostkeycheck

B. Router(config)#ip domain-name cisco.com
Router(config)#crypto key generate rsa general-keys modulus 1024
Router(config)#ip ssh version 2
Router(config-line)#line vty 0 15
Router(config-line)# transport input all
Router(config)#ip ssh logging events

C. Router(config)#hostname R15
R15(config)#ip domain-name cisco.com
R15(config)#crypto key generate rsa general-keys modulus 1024
R15(config)#ip ssh version 2
R15(config-line)#line vty 0 15
R15(config-line)# transport input ssh

D. Router(config)#crypto key generate rsa general-keys modulus 1024
Router(config)#ip ssh version 2
Router(config-line)#line vty 0 15
Router(config-line)# transport input ssh
Router(config)#ip ssh logging events
R15(config)#ip ssh stricthostkeycheck

Answer: C

Explanation

Steps to configure SSH:

1. Configure the router hostname using command “hostname”.
2. Configure the domain name using command “ip domain-name”.
3. Generate public and private keys using command “crypto key generate rsa”.
4. Create a user in the local database using command “username...secret”.
5. Allow only SSH access on VTY lines using command “transport input ssh”.

Reference: <https://ipwithease.com/how-to-configure-ssh-version-2-on-cisco-router/>

Question 41

Refer to the exhibit.

The screenshot shows the configuration interface for a Cisco Wireless LAN Controller. The 'Fast Transition' section at the top has a 'Disable' button. Below it, the 'Protected Management Frame' section has a 'PMF' setting set to 'Disabled'. The 'WPA+WPA2 Parameters' section includes checkboxes for 'WPA Policy' (unchecked), 'WPA2 Policy' (checked), 'WPA2 Encryption' (checked), and 'OSN Policy' (unchecked). Under 'WPA2 Encryption', there are radio buttons for 'AES' (selected), 'TKIP', 'CCMP256', 'GCMP128', and 'GCMP256'. The 'Authentication Key Management' section at the bottom has checkboxes for '802.1X' (unchecked), 'CCKM' (unchecked), 'PSK' (checked), 'FT 802.1X' (unchecked), and 'FT PSK' (unchecked).

Users need to connect to the wireless network with IEEE 802.11r-compatible devices. The connection must be maintained as users travel between floors or to other areas in the building. What must be the configuration of the connection?

- A. Select the WPA Policy option with the CCKM option
- B. Disable AES encryption
- C. Enable Fast Transition and select the FT 802.1x option
- D. Enable Fast Transition and select the FT PSK option

Answer: C

Explanation

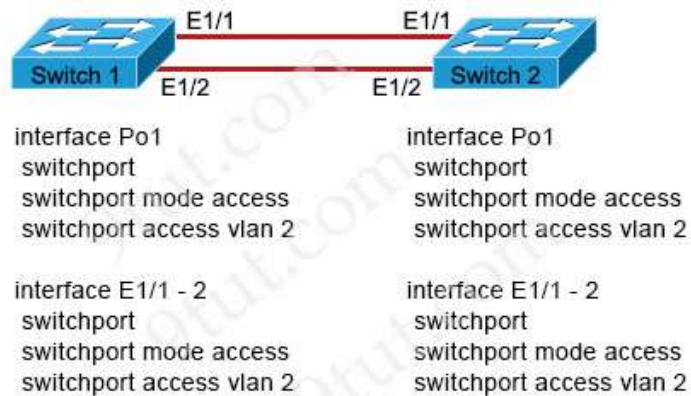
802.11r, which is the IEEE standard for fast roaming, introduces a new concept of roaming where the initial handshake with the new AP is done even before the client roams to the target AP, which is called Fast Transition (FT). The initial handshake allows the client and APs to do the Pairwise Transient Key (PTK) calculation in advance. These PTK keys are applied to the client and AP after the client does the reassociation request or response exchange with new target AP.

FT works with both preshared key (PSK) and 802.1X authentication methods. If you check the FT PSK check box, from the PSK Format drop-down list, choose ASCII or Hex and enter the key value -> This question does not mention about "enter the key value" so maybe answer C is the best choice.

Reference: <https://www.cisco.com/c/dam/en/us/td/docs/wireless/controller/technotes/80211r-ft/b-80211r-dg.html>

Question 42

Refer to the exhibit.



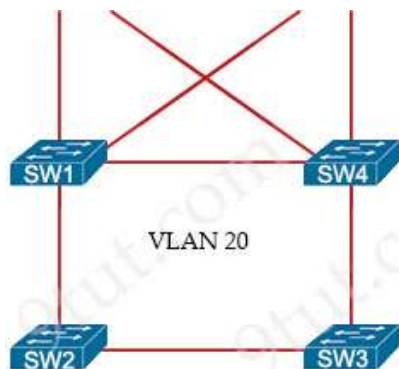
An engineer is configuring an EtherChannel using LACP between Switches 1 and 2. Which configuration must be applied so that only Switch 1 sends LACP initiation packets?

- A. Switch1(config-if)#channel-group 1 mode on
Switch2(config-if)#channel-group 1 mode passive
- B. Switch1(config-if)#channel-group 1 mode passive
Switch2(config-if)#channel-group 1 mode active
- C. Switch1(config-if)#channel-group 1 mode active
Switch2(config-if)#channel-group 1 mode passive
- D. Switch1(config-if)#channel-group 1 mode on
Switch2(config-if)#channel-group 1 mode active

Answer: C

Question 43

Refer to the exhibit.



```
SW1 = 24596 0018.184e.3c00
SW2 = 28692 004a.14e5.4077
SW3 = 32788 0022.55cf.dd00
SW4 = 64000 0041.454d.407f
```

Which switch becomes the root of a spanning tree for VLAN 20 if all links are of equal speed?

- A. SW1
- B. SW2
- C. SW3
- D. SW4

Answer: A

Question 44

Refer to the exhibit.



```
Router1(config)#interface GigabitEthernet0/0
Router1(config-if)#ip address 209.165.200.225 255.255.255.224
Router1(config-if)#ip nat outside
Router1(config)#interface GigabitEthernet0/1
Router1(config-if)#ip nat inside
Router1(config)#interface GigabitEthernet
Router1(config-if)#encapsulation dot1Q 100
Router1(config-if)#ip address 10.10.10.1 255.255.255.0
Router1(config)#interface GigabitEthernet0/1.200
Router1(config-if)#encapsulation dot1Q 200
Router1(config-if)#ip address 10.10.20.1 255.255.255.0
Router1(config)#ip access-list standard NAT_INSIDE_RANGES
Router1(config-std-nacl)#permit 10.10.10.0 0.0.0.255
Router1(config)#ip nat inside source list NAT_INSIDE_RANGES interface GigabitEthernet0/0 overload
```

Users on existing VLAN 100 can reach sites on the Internet. Which action must the administrator take to establish connectivity to the Internet for users in VLAN 200?

- A. Define a NAT pool on the router.
- B. Update the NAT_INSIDE_RANGES ACL
- C. Configure the **ip nat outside** command on another interface for VLAN 200
- D. Configure static NAT translations for VLAN 200

Answer: B

Explanation

We need to add the “permit 10.10.20.0 0.0.0.255” command to the ACL.

Question 45

Which protocol uses the SSL?

- A. HTTP
- B. HTTPS
- C. SSH
- D. Telnet

Answer: B

Question 46

Drag and drop the facts about wireless architectures from the left onto the types of access point on the right. Not all options are used.

configured and managed by a WLC	Autonomous Access Point
requires a management IP address	
managed from a web-based dashboard	Cloud-Based Access Point
supports automatic deployment	
accessible for management via Telnet, SSH, or a web GUI	

Answer:

Autonomous Access Point

- + requires a management IP address
- + accessible for management via Telnet, SSH, or a web GUI

Cloud-Based Access Point

- + managed from a web-based dashboard
- + supports automatic deployment

Explanation

An **autonomous AP** is a self-contained device with both wired and wireless hardware so that it can bridge to the wired VLAN infrastructure wireless clients that belong to SSIDs. Each autonomous AP must be configured with a management IP address so that it can be remotely accessed using Telnet, SSH, or a web interface. Each AP must be individually managed and maintained unless you use a management platform such as Cisco DNA Center.

Cloud-based AP management is an alternative to purchasing a management platform. The AP management function is pushed into the Internet cloud. For example, Cisco Meraki is a cloud-based AP management service that allows you to automatically deploy Cisco Meraki APs. These APs can then be managed from the Meraki cloud web interface (dashboard).

Reference: <https://www.ciscopress.com/articles/article.asp?p=2999384&seqNum=5>

From the above paragraph, we can see autonomous AP is not managed by a WLC.

Question 47

Which value is the unique identifier that an access point uses to establish and maintain wireless connectivity to wireless network devices?

- A. VLANID
- B. SSID
- C. RFID
- D. WLANID

Answer: B

Explanation

The SSID is a unique identifier that wireless networking devices use to establish and maintain wireless connectivity. Multiple access points on a network or subnetwork can use the same SSIDs. SSIDs are case sensitive and can contain up to 32 alphanumeric characters.

Question 48

A network engineer is configuring a switch so that it is remotely reachable via SSH. The engineer has already configured the host name on the router. Which additional command must the engineer configure before entering the command to generate the RSA key?

- A. password password
- B. crypto key generate rsa modulus 1024
- C. ip domain-name domain
- D. ip ssh authentication-retries 2

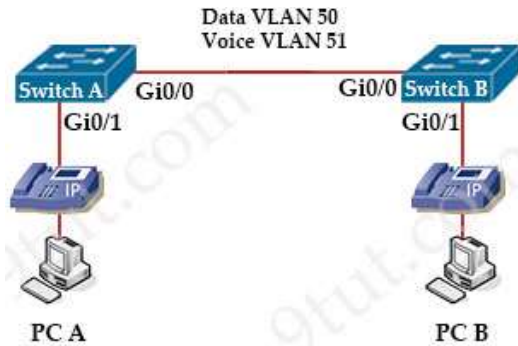
Answer: C

Explanation

The command “crypto key generate rsa modulus 1024” is used to generate the RSA key but this question asks about the command which is typed before entering the command to generate the RSA key so answer C is the best choice.

Question 49

Refer to the exhibit.



Switch A is newly configured. All VLANs are present in the VLAN database. The IP phone and PC A on Gi0/1 must be configured for the appropriate VLANs to establish connectivity between the PCs. Which command set fulfills the requirement?

- A. SwitchA(config-if)#switchport mode access
SwitchA(config-if)#switchport access vlan 50
SwitchA(config-if)#switchport voice vlan 51
- B. SwitchA(config-if)#switchport mode access
SwitchA(config-if)#switchport access vlan 50
SwitchA(config-if)#switchport voice vlan untagged
- C. SwitchA(config-if)#switchport mode trunk
SwitchA(config-if)#switchport trunk allowed vlan add 50, 51
SwitchA(config-if)#switchport voice vlan dot1p
- D. SwitchA(config-if)#switchport mode trunk
SwitchA(config-if)#switchport trunk allowed vlan 50, 51
SwitchA(config-if)#switchport qos trust cos

Answer: A

Question 50

Which QoS traffic handling technique retains excess packets in a queue and reschedules these packets for later transmission when the configured maximum bandwidth has been surpassed?

- A. traffic shaping
- B. traffic policing
- C. weighted random early detection
- D. traffic prioritization

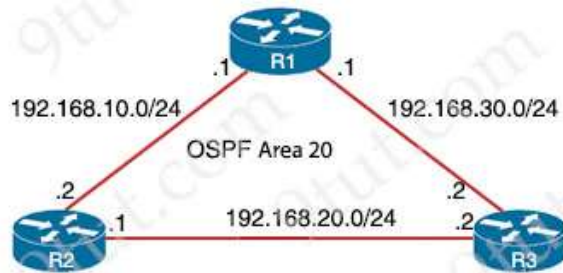
Answer: A

Explanation

+ **Shaping**: retains excess packets in a queue and then schedules the excess for later transmission over increments of time. When traffic reaches the maximum configured rate, additional packets are queued instead of being dropped to proceed later. Traffic shaping is applicable only on outbound interfaces as buffering and queuing happens only on outbound interfaces. Shaping is configured in bits per second.

Question 51

Refer to the exhibit.



R1 learns all routes via OSPF. Which command configures a backup static route on R1 to reach the 192.168.20.0/24 network via R3?

- A. R1(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.2 111
- B. R1(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.2 90
- C. R1(config)#ip route 192.168.20.0 255.255.0.0 192.168.30.2
- D. R1(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.2

Answer: A

Question 52

Which Layer 2 switch function encapsulates packets for different VLANs so that the packets traverse the same port and maintain traffic separation between the VLANs?

- A. VLAN numbering
- B. VLAN DSCP
- C. VLAN tagging
- D. VLAN marking

Answer: C

Explanation

VLAN tagging is a method through which more than one VLAN is handled on a port. VLAN tagging is used to tell which packet belongs to which VLAN on the other side. To make recognition easier, a packet is tagged with a VLAN tag in the Ethernet frame.

Question 53

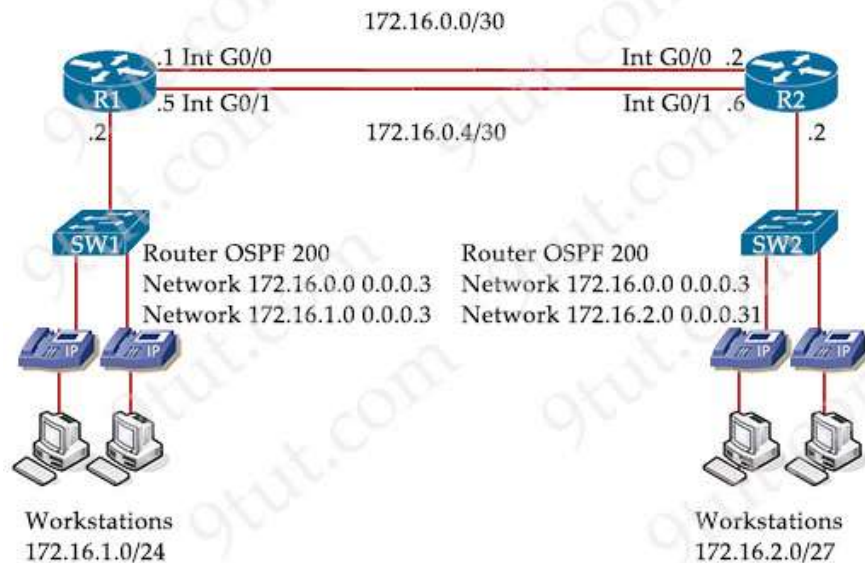
What is an expected outcome when network management automation is deployed?

- A. A distributed management plane must be used.
- B. Software upgrades are performed from a central controller
- C. Complexity increases when new device configurations are added
- D. Custom applications are needed to configure network devices

Answer: B

Question 54

Refer to the exhibit.



The primary route across Gi0/0 is configured on both routers. A secondary route must be configured to establish connectivity between the workstation networks. Which command set must be configured to complete this task?

A. R1
ip route 172.16.2.0 255.255.255.240 172.16.0.2 113

R2
ip route 172.16.1.0 255.255.255.0 172.16.0.1 114

B. R1
ip route 172.16.2.0 255.255.255.240 172.16.0.5 89

R2
ip route 172.16.1.0 255.255.255.0 172.16.0.6 89

C. R1
ip route 172.16.2.0 255.255.255.248 172.16.0.5 110

R2
ip route 172.16.1.0 255.255.255.0 172.16.0.6 110

D. R1
ip route 172.16.2.0 255.255.255.224 172.16.0.6 111

R2
ip route 172.16.1.0 255.255.255.0 172.16.0.5 112

Answer: D

===== New Questions (added on 4th-Mar-2022) =====

Question 55

Refer to the exhibit.

```
Cat9300#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
```

Which action must be taken so that neighboring devices rapidly discover switch Cat9300?

- A. Enable portfast on the ports that connect to neighboring devices
- B. Configure the **cdp holdtime 10** command on switch Cat9300
- C. Configure the **cdp timer 10** command on the neighbors of switch Cat9300
- D. Configure the **cdp timer 10** command on switch Cat9300

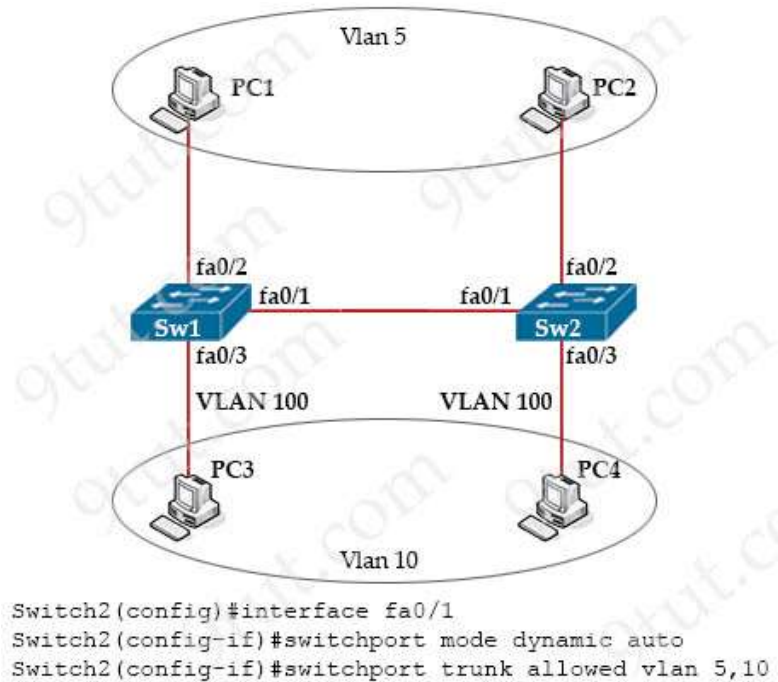
Answer: D

Explanation

The “**cdp timer seconds**” command sets the transmission frequency of CDP updates in seconds so we should reduce it so that CDP updates are sent more often.

Question 56

Refer to the exhibit.



SW2 is replaced due to a hardware failure. A network engineer starts to configure SW2 by copying the Fa0/1 interface configuration from SW1. Which command must be configured on the Fa0/1 interface of SW2 to enable PC1 to connect to PC2?

- A. switchport mode trunk
- B. switchport trunk native vlan 10
- C. switchport trunk allowed remove 10
- D. switchport mode access

Answer: A

===== New Questions (added on 11th-Mar-2022) =====

Question 57

How do UTP and STP cables compare?

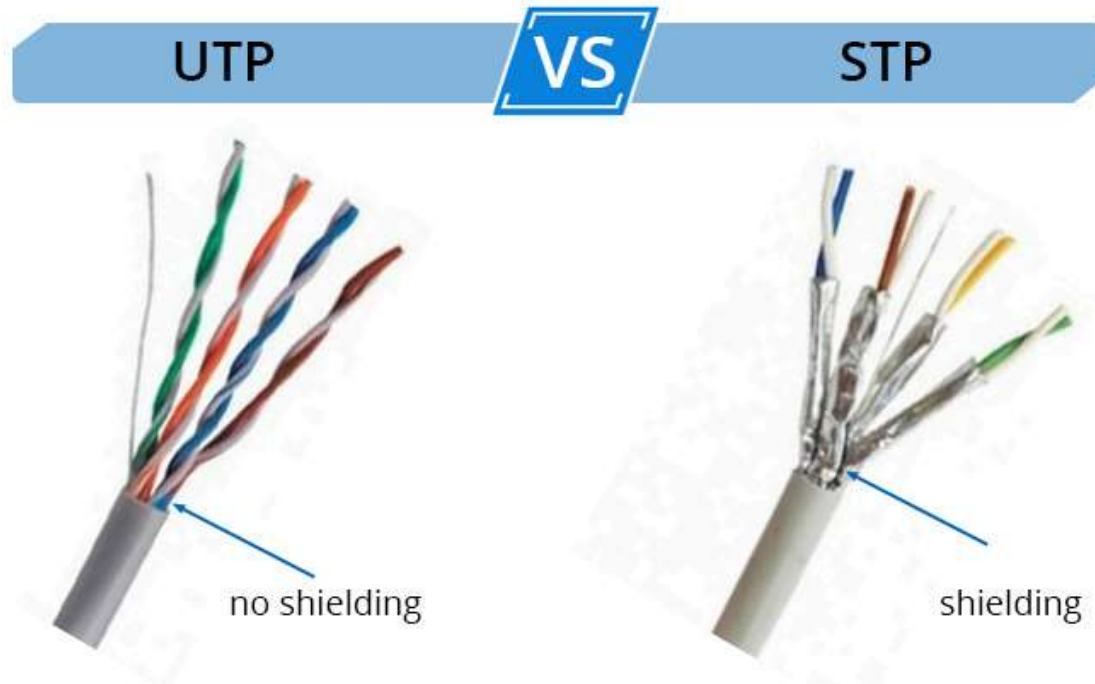
- A. STP cables are cheaper to produce and easier to install and UTP cables are more expensive and harder to install.
- B. UTP cables are less prone to crosstalk and interference and STP cables are more prone to crosstalk and interference.
- C. UTP cables provide faster and more reliable data transfer rates and STP cables are slower and less reliable.
- D. STP cables are shielded and protect against electromagnetic interference and UTP lacks the same protection against electromagnetic interference.

Answer: D

Explanation

UTP (Unshielded twisted pair) and STP (Shielded twisted pair) are the types of twisted pair cables which act as a transmission medium and imparts reliable connectivity of electronic equipment. Although the design and manufacture are different but both serve the same purpose.

The basic difference between UTP and STP is UTP (Unshielded twisted pair) is a cable with wires that are twisted together to reduce noise and crosstalk. On the contrary, STP (Shielded twisted pair) is a twisted pair cable confined in foil or mesh shield that guards the cable against electromagnetic interference.



Question 58

Drag and drop the statements about device management from the left onto the corresponding device-management types on the right.

It uses multiple tools and applications to analyze and troubleshoot different types of data	Cisco DNA Center Device Management
It manages device configurations on a per-device basis	
It provides a single interface for network security and analytics.	
It supports CLI templates to apply a consistent configuration to multiple devices	
Security is managed near the perimeter of the network with firewalls, VPNs, and IPS	Traditional Device Management
It uses NetFlow to analyze potential security threats and take appropriate action on that traffic	

Answer:

Cisco DNA Center Device Management:

- + It provides a single interface for network security and analytics
- + It supports CLI templates to apply a consistent configuration to multiple devices
- + It uses NetFlow to analyze potential security threats and take appropriate action on that traffic

Traditional Device Management:

- + It uses multiple tools and applications to analyze and troubleshoot different types of data
- + It manages device configurations on a per-device basis
- + Security is managed near the perimeter of the network with firewalls, VPNs, and IPS

===== New Questions (added on 15th-Mar-2022) =====

Question 59

Which port type does a lightweight AP use to connect to the wired network when configured in FlexConnect mode with local switching and VLAN tagging?

- A. EtherChannel
- B. access
- C. LAG
- D. trunk

Answer: D

Explanation

Local Switched: Locally-switched WLAN's (the SSID you are connected to) will map their wireless user traffic to a VLAN via 802.1Q trunking to a local switch adjacent to the access point.

Reference: <https://wlanlessonslearned.wordpress.com/tag/flexconnect/>

Question 60

An engineer is installing a new wireless printer with a static IP address on the Wi-Fi network. Which feature must be enabled and configured to prevent connection issues with the printer?

- A. passive client
- B. static IP tunneling
- C. DHCP address assignment
- D. client exclusion

Answer: A

Explanation

Passive clients are wireless devices, such as scales and printers that are configured with a static IP address. These clients do not transmit any IP information such as IP address, subnet mask, and gateway information when they associate with an access point. As a result, when passive clients are used, the controller never knows the IP address unless they use the DHCP.

Since the wireless controller does not have any IP related information about passive clients, it cannot respond to any ARP requests. The current behavior does not allow the transfer of ARP requests to passive clients. Any application that tries to access a passive client will fail.

The **passive client feature enables the ARP requests and responses to be exchanged between wired and wireless clients**. This feature when enabled, allows the controller to pass ARP requests from wired to wireless clients until the desired wireless client gets to the RUN state.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-6/configuration-guide/b_cg76/b_cg76_chapter_01100000.pdf

Question 61

An engineer is configuring router R1 with an IPv6 static route for prefix 2019:C15C:0CAF:E001::/64. The next hop must be 2019:C15C:0CAF:E002::1. The route must be reachable via the R1 Gigabit 0/0 interface. Which command configures the designated route?

- A. R1(config)#ipv6 route 2019:C15C:0CAF:E001::/64 2019:C15C:0CAF:E002::1
- B. R1(config-if)#ipv6 route 2019:C15C:0CAF:E001::/64 2019:C15C:0CAF:E002::1
- C. R1(config-if)#ip route 2019:C15C:0CAF:E001::/64 GigabitEthernet0/0
- D. R1(config)#ip route 2019:C15C:0CAF:E001::/64 GigabitEthernet0/0

Answer: A

Question 62

Refer to the exhibit. What must be configured to enable 802.11w on the WLAN?

The screenshot shows the 'Security' tab of a WLAN configuration interface. Under the 'Layer 2' sub-tab, the 'Layer 2 Security' dropdown is set to 'WPA+WPA2'. The 'Security Type' is 'Enterprise'. 'MAC Filtering' is disabled. Under 'WPA+WPA2 Parameters', 'WPA Policy' is disabled, 'WPA2 Policy' is checked, 'WPA2 Encryption' is checked with 'CCMP128(AES)' selected, and 'OSN Policy' is disabled. Under 'Fast Transition', 'Fast Transition' is set to 'Disabled'. Under 'Protected Management Frame', 'PMF' is set to 'Disabled'. Under 'Authentication Key Management', '802.1X-SHA1' is checked and 'Enable' is selected.

- A. Set PMF to Required
- B. Enable MAC Filtering
- C. Enable WPA Policy
- D. Set Fast Transition to Enabled

Answer: A

Explanation

Configuring 802.11w (GUI)

Procedure

Step 1 Choose WLANs > WLAN ID to open the WLANs > Edit page.

Step 2 In the Security tab, choose the Layer 2 security tab.

Step 3 From the Layer 2 Security drop-down list, choose WPA+WPA2.

The 802.11w IGTK Key is derived using the 4-way handshake, which means that it can only be used on WLANs that are configured for WPA2 security at Layer 2.

Note: WPA2 is mandatory and encryption type must be AES. TKIP is not valid.

Step 4 Choose the PMF state from the drop-down list

The following options are available:

Disabled—Disables 802.11w MFP protection on a WLAN

Optional—To be used if the client supports 802.11w.

Required—Ensures that the clients that do not support 802.11w cannot associate with the WLAN.

Reference: https://content.cisco.com/chapter.sjs?uri=/searchable/chapter/content/en/us/td/docs/wireless/controller/7-5/configuration-guide/b_cg75/b_cg75_chapter_01001100.html.xml

Note: When you create a MAC address filter on WLCs, users are granted or denied access to the WLAN network based on the MAC address of the client they use

Fast Transition is 802.11r, not 802.11w.

Therefore “Set PMF to Required” is the best choice.

Question 63

Drag and drop the IPv6 address details from the left onto the corresponding types on the right.

	Anycast
used exclusively by a non-host device	
includes link-local and loopback addresses	
identifies an interface on an IPv6 device	Multicast
assigned to more than one interface	
derived from the FF00::/8 address range	
provides one-to-many communications	Unicast

Answer:

Anycast:

- + used exclusively by a non-host device
- + assigned to more than one interface

Multicast:

- + derived from the FF00::/8 address range
- + provides one-to-many communications

Unicast:

- + includes link-local and loopback addresses
- + identifies an interface on an IPv6 device

Explanation

An anycast address must be assigned to a router not a host and cannot be used as a source address.

===== New Questions (added on 28th-Mar-2022) =====

Question 64

Drag and drop the elements of a security program from the left onto the corresponding descriptions on the right.

awareness	tactical document that sets out specific tasks and methods to maintain security
education	user-awareness learning level that focuses on security practice that all employees must understand and enforce
security policy	user-awareness learning level that focuses on teaching employees how to perform tasks specifically required by their jobs
security standard	document that outlines an organization's security goals and practices and the roles and responsibilities of the organization's personnel
training	user-awareness learning level that focuses on learning about topics and practices beyond what is typically required by the user's job

Answer:

- + **awareness:** user-awareness learning level that focuses on security practice that all employees must understand and enforce
- + **education:** user-awareness learning level that focuses on learning about topics and practices beyond what is typically required by the user's job
- + **security policy:** document that outlines an organization's security goals and practices and the roles and responsibilities of the organization's personnel
- + **security standard:** tactical document that sets out specific tasks and methods to maintain security
- + **training:** user-awareness learning level that focuses on teaching employees how to perform tasks specifically required by their jobs

Explanation

As per Cisco, an effective security program has three basic elements. First, user awareness. Second, user training. And last, physical access control.

User awareness: All users should be made aware of the need for data confidentiality to protect corporate information, as well as their own credentials and personal information.

User training: All users should be required to participate in periodic formal training so that they become familiar with all corporate security policies.

Reference: CCNA 200-301 Official Cert Guide

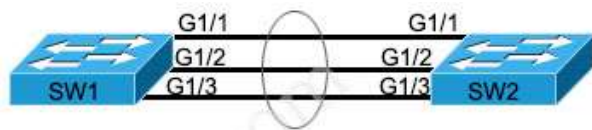
Security Policy: Policies are the top tier of formalized security documents. These high-level documents offer a general statement about the organization's assets and what level of protection they should have.

Standards are much more specific than policies. Standards are **tactical documents** because they lay out specific steps or processes required to meet a certain requirement.

Reference: <https://www.pearsonitcertification.com/articles/article.aspx?p=418007&seqNum=5>

Question 65

Refer to the exhibit.



Which configuration establishes a Layer 2 LACP EtherChannel when applied to both switches?

A. interface range G1/1 -1/3
switchport mode access
channel-group 1 mode passive
no shutdown

B. interface range G1/1 -1/3
switchport mode trunk
channel-group 1 mode desirable
no shutdown

C. interface range G1/1 -1/3
switchport mode trunk
channel-group 1 mode active
no shutdown

D. interface range G1/1 -1/3
switchport mode access
channel-group 1 mode on
no shutdown

Answer: C

Explanation

The question clear stated that we use LACP so we must use mode “active” on both interfaces.

Question 66

Refer to the exhibit.

Security

- AAA
 - General
 - RADIUS
 - Authentication
 - Accounting
 - Fallback
 - DNS
 - Downloaded AVP
 - TACACS+
 - LDAP
 - Local Net Users
 - MAC Filtering
 - Disabled Clients
 - User Login Policies
 - AP Policies
 - Password Policies
 - Local EAP
 - Advanced EAP
 - Priority Order
 - Certificate
 - Access Control Lists
 - Wireless Protection Policies

RADIUS Authentication Servers > New

Server Index (Priority): 2

Server IP Address (IPv4/IPv6): 192.168.25.2

Shared Secret Format: ASCII

Shared Secret:

Confirm Shared Secret:

Key Wrap: ☐ (Designed for FIPS customers and requires a key wrap compliant RADIUS server)

Port Number: 1812

Server Status: Enabled

Support for CoA: Disabled

Server Timeout: 2 seconds

Network User: ☐ Enable

Management: ☐ Enable

Management Retransmit Timeout: 2 seconds

Tunnel Proxy: ☐ Enable

IPSec: ☐ Enable

A network engineer configures the Cisco WLC to authenticate local wireless clients against a RADIUS server. Which task must be performed to complete the process?

- A. Disable the Server Status option
- B. Enable the Management option
- C. Enable the Network User option
- D. Enable the Support for CoA option

Answer: C

Explanation

If you are configuring a RADIUS authentication server, check the **Management** check box to enable management authentication (authenticate users who login to the the WLC). If you enable this feature, this entry is considered the RADIUS authentication server for management users, and authentication requests go to the RADIUS server.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-5/config-guide/b_cg85/aaa_administration.html

The **Network User** check box to enable network user authentication. So the best answer here should be enable “Network User” as it also authenticates local users.

Note:

- + **Support for CoA** (change of authorization) is an extension to the RADIUS protocol that allows dynamic changes to a user session.
- + The **Server Status** checkbox is used to enable/disable this RADIUS server.

Question 67

Which QoS queuing method discards or marks packets that exceed the desired bit rate of traffic flow?

- A. shaping
- B. policing
- C. CBWFQ
- D. LLQ

Answer: B

Explanation

Policing: is used to control the rate of traffic flowing across an interface. During a bandwidth exceed (crossed the maximum configured rate), the excess traffic is generally dropped or remarked.

Question 68

What is the role of disaggregation in controller-based networking?

- A. It divides the control-plane and data-plane functions.
- B. It summarizes the routes between the core and distribution layers of the network topology.
- C. It enables a network topology to quickly adjust from a ring network to a star network
- D. It streamlines traffic handling by assigning individual devices to perform either Layer 2 or Layer 3 functions.

Answer: A

Explanation

Taken literally, “network disaggregation” means to separate the network into its component parts.

Question 69

Which REST method updates an object in the Cisco DNA Center Intent API?

- A. CHANGE
- B. UPDATE
- C. POST
- D. PUT

Answer: D

Explanation

The RESTful Cisco DNA Center Intent API uses HTTPS verbs (GET, POST, PUT, and DELETE) with JSON structures to discover and control the network.

Reference: <https://developer.cisco.com/docs/dna-center/#!cisco-dna-center-platform-overview/intent-api-northbound>

Question 70

Drag and drop the QoS terms from the left onto the descriptions on the right.

Categorizes packets based on the value of traffic descriptor	Class-based weighted fair queuing
Outcome of overutilization	Classification
Guarantees minimum bandwidth to specific traffic classes when an interface is congested	Congestion
Uses defined criteria to limit the transmission of one or more classes of traffic	Policing
Prevents congestion by reducing the flow of the outbound traffic	Shaping

Answer:

Class-based weighted fair queuing: Guarantees minimum bandwidth to specific traffic classes when an interface is congested

Classification: Categorizes packets based on the value of traffic descriptor

Congestion: Outcome of overutilization

Policing: Uses defined criteria to limit the transmission of one or more classes of traffic

Shaping: Prevents congestion by reducing the flow of the outbound traffic

Question 71

What is the function of Cisco Advanced Malware protection for next-generation IPS?

- A. authorizing potentially compromised wireless traffic
- B. URL filtering
- C. authenticating end users
- D. inspecting specific files and files types for malware

Answer: D

Question 72

An administrator must use the **password complexity not manufacturer-name** command to prevent users from adding “cisco” as a password. Which command must be issued before this command?

- A. Password complexity enable
- B. confreg 0x2142
- C. login authentication my-auth-list
- D. service password-encryption

Answer: A

Explanation

Step 3. (Optional) To enable the password complexity settings on the switch, enter the following:

SG350X(config)#**passwords complexity enable**

Reference: <https://www.cisco.com/c/en/us/support/docs/smb/switches/cisco-small-business-300-series-managed-switches/smb5563-configure-password-settings-on-a-switch-through-the-command.html>

Question 73

What is the function of “off-the-shelf” switches in a controller-based network?

- A. Forwarding packets
- B. Making routing decision
- C. Providing a central view of the deployed network
- D. Setting packet-handling policies

Answer: A

Explanation

In SDN’s purest form, the controller has all the intelligence: Switches are dumb, commercial off-the-shelf (COTS) devices that are managed by the controllers.

Therefore we can deduce “off-the-shelf” switches are only used to forward packets.

Question 74

Which two practices are recommended for an acceptable security posture in a network? (Choose two)

- A. Maintain network equipment in a secure location
- B. Backup device configurations to encrypted USB drives for secure retrieval

- C. Use a cryptographic keychain to authenticate to network devices
- D. Place internal email and file servers in a designated DMZ
- E. Disable unused or unnecessary ports, interfaces and services

Answer: A E

Question 75

Refer to the exhibit.

The image shows a Cisco IOS configuration window for Layer 2 Security. The 'Layer 2' tab is selected. The 'Layer 2 Security' dropdown is set to 'Fast Transition'. The 'Fast Transition' checkbox is unchecked. The 'Protected Management Frame' (PMF) is set to 'Disabled'. Under 'WPA+WPA2 Parameters', 'WPA Policy' and 'WPA2 Policy' are both unchecked. 'WPA2 Encryption' is set to 'AES'. Under 'Authentication Key Management', '802.1X', 'CCKM', and 'PSK' are all unchecked.

What are the two steps an engineer must take to provide the highest encryption and authentication using domain credentials from LDAP? (Choose two)

- A. Select WPA policy with TKIP Encryption
- B. Select WPA + WPA2 on layer 2 security
- C. Select PSK under authentication key management
- D. Select Static-WEP + 802.1x on Layer 2 security
- E. Select 802.1x from under authentication key management

Answer: B E

===== New Questions (added on 2nd-May-2022) =====

Question 76

Refer to the exhibit.

```
Cat9K-1#show lldp entry Cat9K-2
```

```
Local Intf: Gi1/0/21  
Chassis ID: 308b.b2b3.2880  
Port id: Gi1/0/21  
Port Description: GigabitEthernet1/0/21  
System Name: Cat9K-2
```

```
Management Addresses:  
IP: 10.6.110.2
```

The network administrator must prevent the switch Cat9K-2 IP address from being visible in LLDP without disabling the protocol. Which action must be taken must be taken to complete the task?

- A. Configure the **no lldp tlv-select-management-address** command globally on Cat9K-2
- B. Configure the **no lldp transmit** command on interface G1/0/21 in Cat9K-1
- C. Configure the **no lldp receive** command on interface G1/0/21 on Cat9K-1
- D. Configure the **no lldp mac-phy-cfg** command globally on Cat9K-2

Answer: A

Explanation

LLDP stands for Link Layer Discovery Protocol and runs over data link layer. It is used for network devices to send information about themselves to other directly connected devices on the network. By using LLDP, two devices running different network layer protocols can learn information about each other. A set of attributes are used to discover neighbor devices. These attributes contains type, length, and value descriptions and are referred to TLVs. Details such as port description, system name, system description, system capabilities, and management address can be sent and received on this Managed Switch.

The command **no lldp tlv-select-management-address** disables the Management Address attribute to be sent.

Question 77

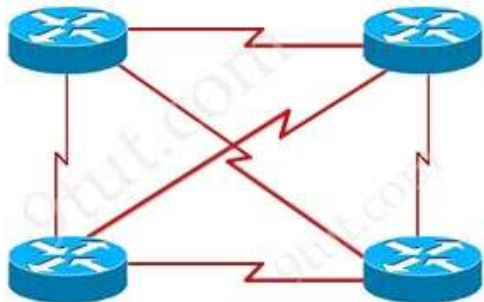
Which WAN topology has the highest degree of reliability?

- A. router-on-a-stick
- B. Point-to-point
- C. hub-and-spoke
- D. full mesh

Answer: D

Explanation

Full-mesh is a network topology in which there is a direct link between all pairs of nodes. Below is an example of full-mesh topology.



Question 78

What is a feature of WPA?

- A. 802.1x authentication
- B. preshared key
- C. TKIP/MIC encryption
- D. small Wi-Fi application

Answer: C

Explanation

WPA uses Michael, a special MIC designed to help with TKIP without requiring excessive computation.

Reference: <https://www.sciencedirect.com/topics/engineering/temporal-key-integrity-protocol>

WPA supports two authentication modes:

- Personal
- Enterprise

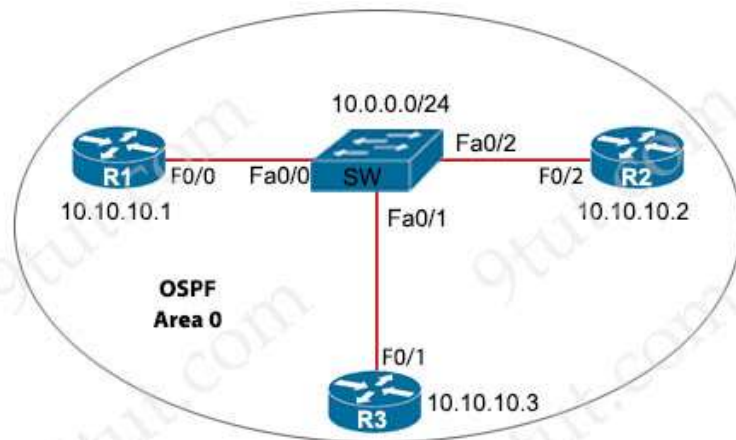
With personal mode, we use a **pre-shared key**. The pre-shared key is not used directly over the air. Instead, wireless clients and the AP use a four-way handshake that uses the pre-shared key as input to generate encryption keys.

Enterprise mode uses 802.1X and an authentication server, usually a RADIUS server.

Therefore in this question, TKIP/MIC encryption is the best choice.

Question 79

Refer to the exhibit.



R1#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.100.2	1	FULL/BDR	00:00:35	192.168.100.2	FastEthernet0/0
192.168.100.3	1	FULL/DR	00:00:34	192.168.100.3	FastEthernet0/0

R1 has taken the DROTHER role in the OSPF DR/BDR election process. Which configuration must an engineer implement so that R1 is elected as the DR?

- A. R1(config)#interface FastEthernet0/0
R1(config-if)#ip ospf priority 1
R1#clear ip ospf process
- B. R1(config)#interface FastEthernet0/0
R1(config-if)#ip ospf priority 200
R1#clear ip ospf process
- C. R3(config)#interface FastEthernet0/1
R3(config-if)#ip ospf priority 200
R3#clear ip ospf process
- D. R2(config)#interface FastEthernet0/2
R2(config-if)#ip ospf priority 1
R2#clear ip ospf process

Answer: B

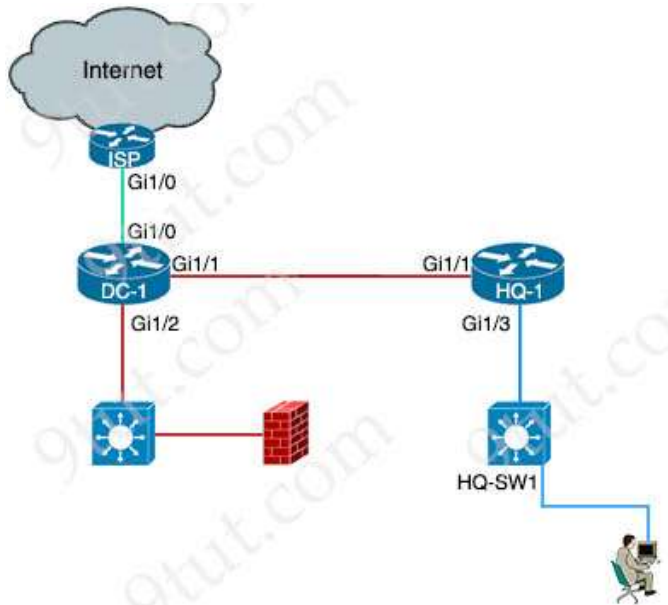
Explanation

From the exhibit we learned that both the priorities of R2 and R3 are 1 so if we set a higher value for R1 on this segment (interface f0/0) and reset the OSPF process than R1 will take the DR role.

===== New Questions (added on 8th-May-2022) =====

Question 80

Refer to the exhibit.



The IP address configurations must be completed on the DC-1 and HQ-1 routers based on these requirements.

DC-1 Gi1/0 must be the last usable address on a /30

DC-1 Gi1/1 must be the first usable address on a /29

DC-1 Gi1/2 must be the last usable address on a /28

HQ-1 Gi1/3 must be the last usable address on a /29

Drag and drop the commands from the left onto the destination interfaces on the right. Not all commands are used.

ip address 192.168.4.13 255.255.255.240	DC-1
ip address 209.165.202.129 255.255.255.252	Gi1/0
ip address 209.165.202.130 255.255.255.252	Gi1/1
ip address 209.165.202.131 255.255.255.252	Gi1/2
ip address 192.168.3.14 255.255.255.240	HQ-1
ip address 192.168.4.9 255.255.255.248	Gi1/3
ip address 192.168.3.14 255.255.255.248	

Answer:

DC-1:

- + ip address 209.165.202.130 255.255.255.252
- + ip address 192.168.4.9 255.255.255.248
- + ip address 192.168.3.14 255.255.255.240

HQ-1:

- + ip address 192.168.3.14 255.255.255.248

Question 81

Refer to the exhibit.

```
cisco_ospf_vrf {"R1 default":  
  ensure => 'present',  
  auto_cost => '100',  
}
```

Which type of configuration is represented in the output?

- A. Chef
- B. JSON
- C. Ansible
- D. Puppet

Answer: D

Explanation

Puppet resource declarations have the following syntax:

```
<TYPE> { '<TITLE>':  
  <ATTRIBUTE> => <VALUE>,  
  <ATTRIBUTE> => <VALUE>,  
}
```

Where:

<TYPE>: is one of the numerous resource types, e.g., file

<TITLE>: A arbitrary unique resource identifier across all resources of the same type (across all resources applied to a particular system)

<ATTRIBUTE>: Resource type-specific parameters

Question 82

How does TFTP operate in a network?

- A. relies on the well-known TCP port 20 to transmit data
- B. requires two separate connections for control and data traffic
- C. uses block numbers to identify and mitigate data-transfer errors
- D. provides secure data transfer

Answer: C

Explanation

TFTP is a simple protocol for transferring files, implemented on top of the UDP/IP protocols using well-known port number 69.

Each file block has an identification number so that ACK messages from target devices can correctly report back to sending devices if a block is received successfully.

Reference: <https://www.ipswitch.com/blog/what-is-tftp-trivial-file-transfer-protocol>

Question 83

Drag and drop the statements about networking from the left onto the corresponding networking types on the right.

This type implements changes individually at each device	Traditional Networking
This type provides a centralized view of the network	
This type leverages controllers to handle network management	
Maintenance costs are higher than with other networking options	Controller-Based Networking

Answer:

Traditional Networking

- + This type implements changes individually at each device.
- + Maintenance costs are higher than with other networking options.

Controller-Based Networking

- + This type provides a centralized view of the network.
- + This type leverages controllers to handle network management.

Question 84

Refer to the exhibit.

```
{  
  "Routers": ["R1", "R2", "R3"],  
  "Switches": ["SW1", "SW2", "SW3"]  
}
```

What is represented by “R1” and “SW1” within the JSON output?

- A. array
- B. object
- C. value
- D. key

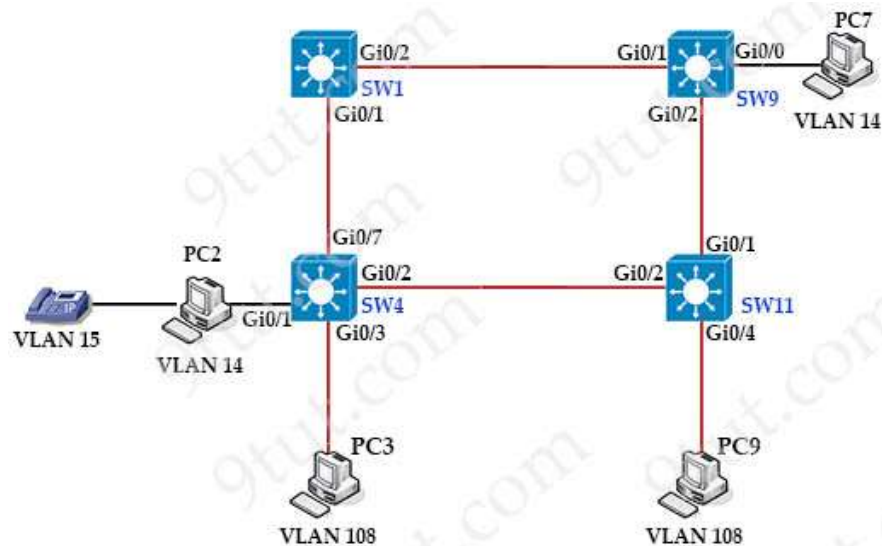
Answer: C

Explanation

In this JSON output, “Routers” and “Switches” are keys while “R1”, “R2”, “R3”, “SW1”, “SW2”, “SW3” are values.

Question 85

Refer to the exhibit.



The following must be considered

- + SW1 is fully configured for all traffic
- + The SW4 and SW9 links to SW1 have been configured
- + The SW4 interface Gi0/1 and Gi0/0 on SW9 have been configured
- + The remaining switches have had all VLANs added to their VLAN database

Which configuration establishes a successful ping from PC2 to PC7 without interruption to traffic flow between other PCs?

Option A	Option B
<pre>SW4# interface Gi0/2 switchport mode trunk switchport trunk allowed vlan 14,108 SW11# interface Gi0/2 switchport mode trunk switchport trunk allowed vlan 14,108 ! interface Gi0/1</pre>	<pre>SW4 interface Gi0/7 switchport mode trunk switchport trunk allowed vlan 108 ! interface Gi0/2 switchport mode access switchport access vlan 14 SW11# interface Gi0/2 switchport mode trunk</pre>

<pre>switchport mode trunk switchport trunk allowed vlan 14,108 SW9# interface Gi0/2 switchport mode trunk switchport trunk allowed vlan 14</pre>	<pre>switchport trunk allowed vlan 14,108 ! interface Gi0/1 switchport mode trunk switchport trunk allowed vlan 14,108 SW9# interface Gi0/2 switchport mode access switchport access vlan 14</pre>
Option C <pre>SW4 interface Gi0/2 switchport mode trunk switchport trunk allowed vlan 14 SW11# interface Gi0/1 switchport mode trunk switchport trunk allowed vlan 14 SW9# interface Gi0/2 switchport mode trunk switchport trunk allowed vlan 108</pre>	Option D <pre>SW4 interface Gi0/2 switchport mode access switchport access vlan 14 SW11# interface Gi0/2 switchport mode access switchport access vlan 14 ! interface Gi0/0 switchport mode access switchport access vlan 14 ! interface Gi0/1 switchport mode trunk SW9# interface Gi0/2 switchport mode access switchport access vlan 14</pre>

- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer: A

Explanation

Option B is not correct as either the link between SW4 & SW11 or the link between SW9 & SW11 must allow VLAN 108. But Gi0/2 of SW4 and Gi0/2 of SW9 only allow VLAN 14 -> PC3 cannot reach PC9 in VLAN 108.

Option C is not correct as Gi0/2 of SW4 and Gi0/1 of SW11 only allows VLAN 14 -> PC3 cannot reach PC9 in VLAN 108.

Option D has configured Gi0/0 of SW11 but there is no such interface in the topology. Also with option D interface Gi0/2 of SW4 and Gi0/2 of SW9 only allow VLAN 14 -> PC3 cannot reach PC9 in VLAN 108.

Question 86

An engineer is configuring switch SW1 to act as an NTP server when all upstream NTP server connectivity fails. Which configuration must be used?

A. SW1# config t

SW1(config)#ntp server 192.168.1.1

SW1(config)#ntp access-group server accesslist1

B. SW1# config t

SW1(config)#ntp peer 192.168.1.1

SW1(config)#ntp access-group peer accesslist1

C. SW1# config t

SW1(config)#ntp backup

SW1(config)#ntp server 192.168.1.1

D. SW1# config t

SW1(config)#ntp master

SW1(config)#ntp server 192.168.1.1

Answer: D

Question 87

Refer to the exhibit.



An OSPF neighbor relationship must be configured using these guidelines:

- R1 is only permitted to establish a neighbor with R2.
- R1 will never participate in DR elections.
- R1 will use a router-id of 10.1.1.1.

Which configuration must be used?

Option A	Option B
----------	----------

<pre> interface Loopback0 ip address 10.1.1.1 255.255.255.255 interface FastEthernet0/0 ip address 10.100.1.1 255.255.255.252 ip ospf priority 100 ip access-group 102 in router ospf 10 log-adjacency-changes network 10.1.1.1 0.0.0.0 area 0 network 10.100.1.0 0.0.0.3 area 0 ospf router-id 10.1.1.1 access-list 102 permit 88 host 10.100.1.2 host 224.0.0.5 access-list 102 deny 88 any any access-list 102 permit ip any any </pre>	<pre> interface FastEthernet0/0 ip address 10.100.1.1 255.255.255.252 ip ospf priority 0 ip access-group 102 in router ospf 10 log-adjacency-changes network 10.1.1.1 0.0.0.0 area 0 network 10.100.1.0 0.0.0.3 area 0 router-id 10.1.1.1 access-list 102 permit 89 host 10.100.1.2 host 224.0.0.5 access-list 102 deny 89 any any access-list 102 permit ip any any </pre>
Option C <pre> interface Loopback0 ip address 10.1.1.1 255.255.255.255 interface FastEthernet0/0 ip address 10.100.1.1 255.255.255.252 ip ospf priority 0 ip access-group 102 in router ospf 10 log-adjacency-changes network 10.1.1.1 0.0.0.0 area 0 network 10.100.1.0 0.0.0.3 area 0 router-id 10.1.1.1 access-list 102 permit 88 host 10.100.1.2 host 224.0.0.5 access-list 102 deny 88 any any access-list 102 permit ip any any </pre>	Option D <pre> interface FastEthernet0/0 ip address 10.100.1.1 255.255.255.252 ip ospf priority 100 ip access-group 102 in router ospf 10 log-adjacency-changes network 10.1.1.1 0.0.0.0 area 0 network 10.100.1.0 0.0.0.3 area 0 ospf router-id 10.1.1.1 access-list 102 permit 89 host 10.100.1.2 host 224.0.0.5 access-list 102 deny 89 any any </pre>

- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer: B

Explanation

R1 will never participate in DR elections -> Set the OSPF priority to 0 ("ip ospf priority 0") can prevent R1 from participate in DR elections -> Only Option B & Option C are correct.

OSPF is protocol 89 so we must permit it from 10.100.1.2 while blocking from others -> Only Option B is correct.

Question 88

How does authentication differ from authorization?

- A. Authentication verifies the identity of a person accessing a network, and authorization determines what resource a user can access.
- B. Authentication is used to determine what resources a user is allowed to access, and authorization is used to track what equipment is allowed access to the network
- C. Authentication is used to verify a person's identity, and authorization is used to create syslog messages for logins
- D. Authentication is used to record what resource a user accesses, and authorization is used to determine what resources a user can access

Answer: A

Question 89

A user configured OSPF and advertised the Gigabit Ethernet interface in OSPF. By default, to which type of OSPF network does this interface belong?

- A. point-to-multipoint
- B. point-to-point
- C. nonbroadcast
- D. broadcast

Answer: D

Question 90

Refer to the exhibit.

```
ip arp inspection vlan 5-10
interface fastethernet0/1
  switchport mode access
  switchport access vlan 5
```

What is the effect of this configuration?

- A. Egress traffic is passed only if the destination is a DHCP server.
- B. All ingress and egress traffic is dropped because the interface is untrusted.
- C. All ARP packets are dropped by the switch.
- D. The switch discards all ingress ARP traffic with invalid MAC-to-IP address bindings.

Answer: D

Explanation

Dynamic ARP inspection is an ingress security feature; it does not perform any egress checking.

Question 91

Refer to the exhibit.

```
R1# show ip route | begin gateway
Gateway of last resort is not set
  172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
C   172.16.1.0/24 is directly connected, FastEthernet0/0
L   172.16.1.1/32 is directly connected, FastEthernet0/0
EX  172.16.2.0/24 [170/2] via 207.165.200.250, 00:00:23, Serial0/0/0
O   192.168.1.0/24 [110/84437] via 207.165.200.254, 00:00:17, Serial0/0/1
D   192.168.2.0/24 [90/184437] via 207.165.200.254, 00:00:13, Serial0/0/1
E1  192.168.3.0/24 [110/1831437] via 207.165.200.254, 00:00:19, Serial0/0/1
    207.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C   207.165.200.248/30 is directly connected, Serial0/0/0
L   207.165.200.249/32 is directly connected, Serial0/0/0
C   207.165.200.252/30 is directly connected, Serial0/0/1
L   207.165.200.253/32 is directly connected, Serial0/0/1
```

Which prefix did router R1 learn from internal EIGRP?

- A. 192.168.2.0/24
- B. 192.168.1.0/24
- C. 192.168.3.0/24
- D. 172.16.1.0/24

Answer: A

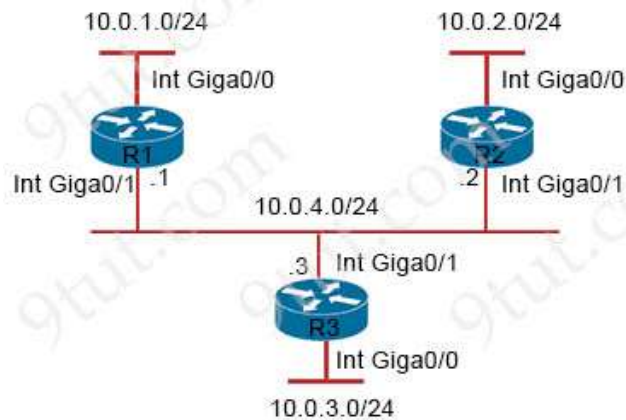
Explanation

Internal EIGRP is symbolized by letter “D”. “EX” means external routes redistributed into EIGRP.

Note: “E1” and “E2” are external OSPF routes.

Question 92

Refer to the exhibit.



Router R1 must be configured to reach the 10.0.3.0 network from the 10.0.1.0/24 segment. Which command must be used to configure the route?

- A. route add 10.0.3.0 0.255.255.255 10.0.4.2
- B. route add 10.0.3.0 mask 255.255.255.0 10.0.4.3
- C. ip route 10.0.3.0 0.255.255.255 10.0.4.2
- D. ip route 10.0.3.0 255.255.255.0 10.0.4.3

Answer: D

Question 93

What is a benefit for external users who consume public cloud resources?

- A. implemented over a dedicated WAN
- B. located in the same data center as the users
- C. all hosted on physical servers
- D. accessed over the Internet

Answer: D

Question 94

In an SDN architecture, which function of a network node is centralized on a controller?

- A. provides protocol access for remote access devices
- B. discards a message due filtering
- C. creates the IP routing table
- D. makes a routing decision

Answer: D

Explanation

The Open Networking Foundation identifies three main parts of the Software-defined networking (SDN): Application layer; Control layer and Infrastructure layer. SDN separates a router's control plane from the data (forwarding) plane. The control plane makes routing decisions. The data plane forwards data (packets) through the router. With SDN routing, decisions are made remotely instead of on each individual router.

Question 95

An engineer must configure neighbor discovery between the company router and an ISP.

```
interface gigabitethernet0/0
description Circuit-ATT4202-89930
duplex full
speed 1000
media-type gbic
negotiation auto
lldp transmit
lldp receive
```

What is the next step to complete the configuration if the ISP uses a third-party router?

- A. Disable CDP on gi0/0.
- B. Disable auto-negotiation.
- C. Enable LLDP TLVs on the ISP router.
- D. Enable LLDP globally.

Answer: D

Question 96

Which SDN plane forwards user-generated traffic?

- A. policy plane
- B. management plane
- C. data plane
- D. control plane

Answer: C

Question 97

When should an engineer implement a collapsed-core architecture?

- A. for large networks that are connected to multiple remote sites
- B. the access and distribution layers must be on the same device
- C. only when using VSS technology
- D. for small networks with minimal need for growth

Answer: D

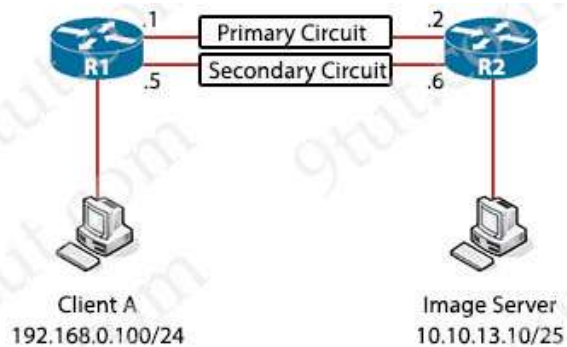
Explanation

The three-tier hierarchical design maximizes performance, network availability, and the ability to scale the network design. Most small enterprise campus' do not grow significantly larger over time, and most small enterprise campus are small enough to be well served by a two-tier hierarchical design, where the core and distribution layers are collapsed into one layer. The primary motivation for the collapsed core design is reducing network cost, while maintaining most of the benefits of the three-tier hierarchical model.

Reference: https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Small_Enterprise_Design_Profile/SEDP/chap2.html

Question 98

Refer to the exhibit.



R1#show ip route

Gateway of last resort is 10.10.10.2 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 10.10.10.2

R2#show ip route

Gateway of last resort is 10.10.10.1 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 10.10.10.1

Routers R1 and R2 have been configured with their respective LAN interfaces. The two circuits are operational and reachable across WAN. Which command set establishes failover redundancy if the primary circuit goes down?

- A. R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.6 2
- R2(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.5 2

B. R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.6

R2(config)#ip route 192.166.0.100 255.255.255.255 10.10.10.5

C. R1(config)#ip route 10.10.13.10 255.255.255.255 10.10.10.2

R2(config)#ip route 192.168.0.100 255.255.255.255 10.10.10.1

D. R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.6

R2(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.5

Answer: A

Question 99

Drag and drop the traffic types from the left onto the QoS delivery mechanisms on the right.

database synchronization traffic	best effort
VoIP traffic	priority queue
standard Web browsing traffic	policing
video traffic	shaping

Answer:

+ best effort: standard Web browsing traffic

+ priority queue: VoIP traffic

+ policing: video traffic

+ shaping: database synchronization traffic

Explanation

+ best effort: standard Web browsing traffic (Regular Web traffic is considered a low priority, TCP or UDP)

+ priority queue: VoIP traffic (VoIP is time sensitive and it uses UDP to speed up the delivery of packets. But it should have priority to guarantee quality)

+ policing: video traffic (Video stream is time sensitive and it uses UDP to speed up the delivery of packets. But the bandwidth for video is much higher than voice so we can drop some packets if there is not enough bandwidth. The quality is still acceptable for most viewers)

+ shaping: database synchronization traffic (DB communications use TCP to warrant the delivery of the packets, queuing the packets don't affect the transmission)

Question 100

What is a function performed by a web server?

- A. provide an application that is transmitted over HTTP
- B. send and retrieve email from client devices
- C. securely store files for FTP access
- D. authenticate and authorize a user's identity

Answer: A

Question 101

Drag and drop the functions of AAA supporting protocols from the left onto the protocols on the right.

	RADIUS
separates all three AAA operations	
encrypts the entire body of the access-request packet	
encrypts only the password when it sends an access request	
Combines authentication and authorization	TACACS+
uses UDP	
uses TCP	

Answer:

RADIUS

- + encrypts only the password when it sends an access request
- + combines authentication and authorization
- + uses UDP

TACACS+

- + separates all three AAA operations
- + encrypts the entire body of the access-request packet
- + uses TCP

Question 102

Which two wireless security standards use Counter Mode Cipher Block Chaining Message Authentication Code Protocol for encryption and data integrity? (Choose two)

- A. WPA2
- B. WPA3
- C. WEP
- D. WPA
- E. Wi-Fi 6

Answer: A B

Explanation

WPA2 mandates the use of a new protocol, counter mode with cipher-block chaining message authentication protocol (CCMP). CCMP uses the AES block cipher, replacing the RC4 cipher used in wired equivalent privacy (WEP) and temporal key integrity protocol (TKIP).

CCMP is a security protocol. It follows carefully designed steps that include the use of the AES specified algorithm to encrypt sensitive data.

CCMP-128 (AES-128 in CCM mode) is used as the minimum encryption algorithm in WPA3-Personal mode.

Question 103

What is a practice that protects a network from VLAN hopping attacks?

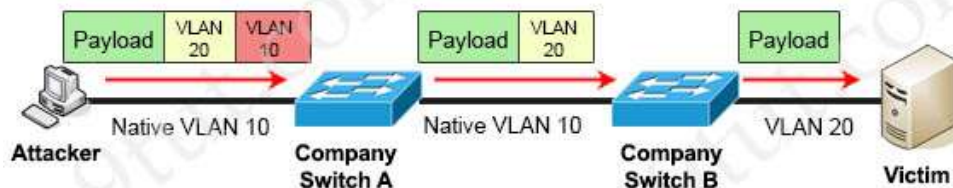
- A. Implement port security on internet-facing VLANs.
- B. Configure an ACL to prevent traffic from changing VLANs.
- C. Assign all access ports to VLANs other than the native VLAN.
- D. Enable dynamic ARP inspection.

Answer: C

Explanation

VLAN Hopping: By altering the VLAN ID on packets encapsulated for trunking, an attacking device can send or receive packets on various VLANs, bypassing Layer 3 security measures. VLAN hopping can be accomplished by switch spoofing or double tagging. One of a popular type of VLAN Hopping is

Double-Tagging attack:



In this attack, the attacking computer generates frames with two 802.1Q tags. The first tag matches the native VLAN of the trunk port (VLAN 10 in this case), and the second matches the VLAN of a host it wants to attack (VLAN 20).

When the packet from the attacker reaches Switch A, Switch A only sees the first VLAN 10 and it matches with its native VLAN 10 so this VLAN tag is removed. Switch A forwards the frame out all links with the same native VLAN 10. Switch B receives the frame with an tag of VLAN 20 so it removes this tag and forwards out to the Victim computer.

Note: This attack only works if the trunk (between two switches) has the same native VLAN as the attacker. In other words, this attack is only successful if the attacker belongs to the native VLAN of the trunk link. Another important point is, this attack is strictly one way as it is impossible to encapsulate the return packet.

To mitigate this type of attack, we can use VLAN access control lists (VACLs, which applies to all traffic within a VLAN. We can use VACL to drop attacker traffic to specific victims/servers); or implement Private VLANs; or keep the native VLAN of all trunk ports different from user VLANs.

Question 104

Refer to the exhibit.

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override
```

Gateway of last resort is 10.56.0.1 to network 0.0.0.0

```
S*      0.0.0.0/0 [1/0] via 10.56.0.1
        10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.56.0.0/17 is directly connected, Vlan56
L        10.56.0.19/32 is directly connected, Vlan56
C        10.56.128.0/18 is directly connected, Vlan57
L        10.56.128.19/32 is directly connected, Vlan57
```

When router R1 is sending traffic to IP address 10.56.192.1, which interface or next hop address does it use to route the packet?

- A. 0.0.0.0/0
- B. Vlan57
- C. 10.56.0.1
- D. 10.56.128.19

Answer: C

Explanation

First we check to see if 10.56.192.1 falls into 10.56.128.0/18 or 10.56.0.0/17 subnets:

10.56.128.0/18 (Increment: 64)
Network address: 10.56.128.0
Broadcast address: 10.56.191.255

10.56.0.0/17 (Increment: 128)
Network address: 10.56.0.0
Broadcast address: 10.56.127.255

As we can see 10.56.192.1 does not belong to above subnets or any other subnets in R1 routing table so R1 will use the default route to forward the packets. This question asks “which interface or next hop address” so the answer is the default route uses the next hop of 10.56.0.1.

Question 105

What is a function of MAC learning on a switch?

- A. A static MAC address is manually added to the MAC table.
- B. MAC address learning is disabled by default on all VLANs.
- C. Frames received for a destination MAC address not listed in the address table are dropped.
- D. The MAC address table is used to populate the ARP table.

Answer: A

Explanation

MAC address learning is enabled by default on all VLANs -> Answer B is not correct.

If a destination MAC address is unknown then the switch will broadcast the frame out all ports except the port where the frame is received -> Answer C is not correct.

The ARP table is built from the replies to the ARP requests, recorded before a packet is sent on the network -> Answer D is not correct.

We can use the “mac address-table static 001d.a18b.36d0 vlan 5 interface fastEthernet 0/2” for example to manually add a MAC address into the MAC table.

Question 106

What is the difference between 1000BASE-LX/LH and 1000BASE-ZX interfaces?

- A. 1000BASE-LX/LH interoperates with multimode and single-mode fiber, and 1000BASE-ZX needs a conditioning patch cable with a multimode.
- B. 1000BASE-ZX is supported on links up to 1000km, and 1000BASE-LX/LH operates over links up to 70 km.
- C. 1000BASE-ZX interoperates with dual-rate 100M/1G 10Km SFP over multimode fiber, and 1000BASE-LX/LH supports only single-rate.
- D. 1000BASE-LX/LH is supported on links up to 10km, and 1000BASE-ZX operates over links up to 70 km.

Answer: D

Explanation

Below lists the cabling standards mentioned above

Standard	Cabling	Maximum length
1000BASE-CX	Twisted pair cabling	25 meters

100BASE-FX	Two strands, multimode	400 m
1000BASE-LX	Long-wavelength laser, MM or SM fiber	10 km (SM) 3 km (MM)
1000BASE-SX	Short-wavelength laser, MM fiber	220 m with 62.5-micron fiber; 550 m with 50-micron fiber
1000BASE-ZX	Extended wavelength, SM fiber	100 km

Note:

- + MM: Multimode
- + SM: Single-mode

Although above list says the maximum length of 1000BASE-ZX is 100km but in practical it is usually from 70 to 80 km depending on link loss.

Question 107

What is the effect when loopback interfaces and the configured router ID are absent during the OSPF Process configuration?

- A. The router ID 0.0.0.0 is selected and placed in the OSPF process.
- B. No router ID is set, and the OSPF protocol does not run.
- C. The highest up/up physical interface IP address is selected as the router ID.
- D. The lowest IP address is incremented by 1 and selected as the router ID.

Answer: C

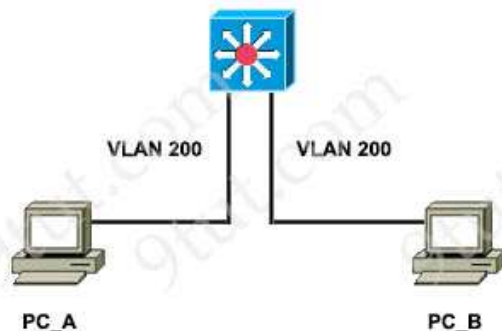
Explanation

OSPF uses the following criteria to select the router ID:

1. Manual configuration of the router ID (via the “router-id x.x.x.x” command under OSPF router configuration mode).
2. Highest IP address on a loopback interface.
3. Highest IP address on a non-loopback and active (no shutdown) interface.

Question 108

Refer to the exhibit.



What is expected when PC_A sends data to PC_B after their initial communication?

- A. The source and destination MAC addresses remain the same
- B. The switch rewrites the source and destination MAC addresses with its own
- C. The source MAC address is changed
- D. The destination MAC address is replaced with ffff.ffff.ffff

Answer: A

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1. noutbuk123123
December 16th, 2024

@9tut,
in question 25 you were wrong, the seventh bit is always inverted, even if there is 1

2. Anonymous
February 20th, 2025

in question 4 answer showing A&C, but option C&D both same , so why not answer D

3. Anonymous
July 23rd, 2025

Can you confirm answer in #39, answer is A but the explanation answer is B. @9tut

4. 9tut
July 24th, 2025

@Anonymous: Thank you for your detection, we updated the explanation of Q39!

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